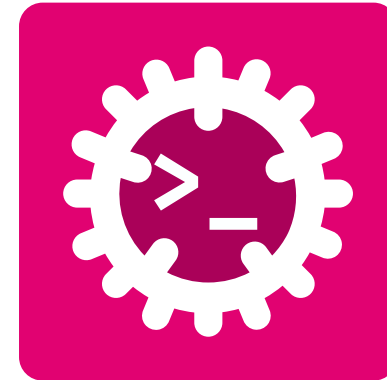


Advanced Programming

Imperative Programming



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2026-08-11 - v0.1.0



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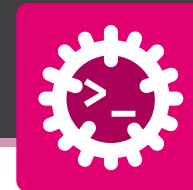


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Imperative Programming

A hands-on, step-by-step approach to coding



During lectures or for small experiments use the [Rust Playground](#) - an online editor with instant compilation and sharing.

```
1 fn main() {  
2  
3     println!("Hello, world!");  
4  
5 }
```

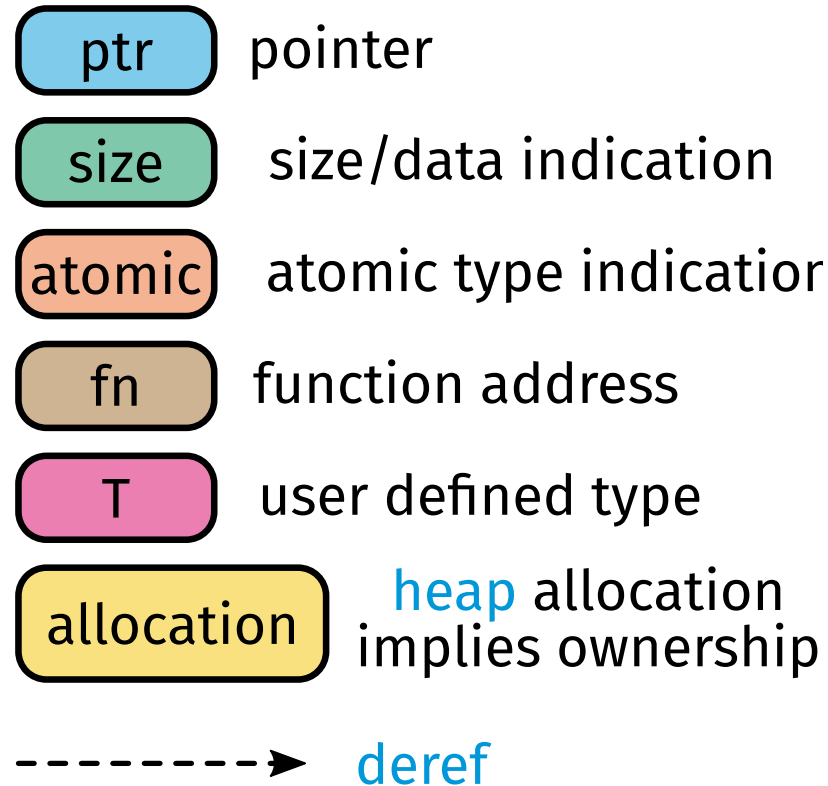
Execution Close

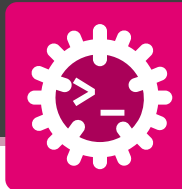
Standard Error

```
Compiling playground v0.0.1 (/playground)  
Finished `dev` profile [unoptimized + debuginfo] target(s) in 1.38s  
Running `target/debug/playground`
```

Standard Output

```
Hello, world!
```





Rust uses different naming styles for different language elements. The compiler accepts many names, but `rustfmt` and `clippy` expect the common Rust style.

Element	Convention	Example
Variable	snake_case	<code>sensor_value</code>
Function	snake_case	<code>read_sensor()</code>
Module	snake_case	<code>sensor_bus</code>
Constant	SCREAMING_SNAKE_CASE	<code>MAX_RETRIES</code>
Static	SCREAMING_SNAKE_CASE	<code>APP_NAME</code>
Type	UpperCamelCase	<code>SensorValue</code>
Struct	UpperCamelCase	<code>TemperatureSensor</code>
Enum	UpperCamelCase	<code>SensorState</code>
Enum variant	UpperCamelCase	<code>OutOfRange</code>
Trait	UpperCamelCase	<code>Readable</code>
Generic type	UpperCamelCase	<code>T</code> , <code>Value</code>

Names are part of the program design.

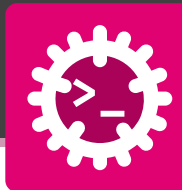


- Good names reduce comments.
- Names should communicate intent.



Constants

Values known at compile time



Definition

A **constant** is a name permanently bound to a **compile-time value**.

- Declared with `const` keyword **anywhere**
- Type annotation is **mandatory**
- Name uses SCREAMING_SNAKE_CASE
- Valid for the **entire** program lifetime
- **Inlined** by the compiler - zero runtime cost

`const` vs `let`

<code>const</code>	<code>let</code>
Compile-time value	Runtime binding
Always immutable	Immutable by default
No memory address	Lives on the Stack
Any scope	Block scope only

```
// Type annotation is required
const MAX_SCORE: u32 = 100_000;
const GRAVITY: f64 = 9.81;
const APP_NAME: &str = "RustApp";
const MAX_RETRIES: u8 = 3;

// Computed constant expression
const SECS_PER_DAY: u64 = 60 * 60 * 24;

fn main() {
    println!("{APP_NAME}");
    println!("max score: {MAX_SCORE}");
    println!("seconds/day: {SECS_PER_DAY}");

    // Constants can be used in other constants
    const DAYS_PER_YEAR: u64 = 365;
    const SECS_PER_YEAR: u64 = SECS_PER_DAY * DAYS_PER_YEAR

    println!("seconds/year: {SECS_PER_YEAR}");
}
```



Min and Max values

Every integer type exposes MIN, MAX and BITS as associated constants:

```
fn main() {
    println!("u8      : {}..={}", u8::BITS, u8::MIN, u8::MAX);
    println!("u16     : {}..={}", u16::BITS, u16::MIN, u16::MAX);
    println!("u32     : {}..={}", u32::BITS, u32::MIN, u32::MAX);
    println!("u64     : {}..={}", u64::BITS, u64::MIN, u64::MAX);
    println!("usize  : {}..={}", usize::MIN, usize::MAX);
    println!("i8      : {}..={}", i8::BITS, i8::MIN, i8::MAX);
    println!("i16     : {}..={}", i16::BITS, i16::MIN, i16::MAX);
    println!("i32     : {}..={}", i32::BITS, i32::MIN, i32::MAX);
    println!("i64     : {}..={}", i64::BITS, i64::MIN, i64::MAX);
    println!("isize  : {}..={}", isize::MIN, isize::MAX);
}
```

Type	Min	Max	Bits
u8	0	255	8
u16	0	65 535	16
u32	0	4.3×10^9	32
u64	0	1.8×10^{19}	64
u128	0	$\approx 3.4 \times 10^{38}$	128
usize	0	platform	32/64
i8	-128	127	8
i16	-32 768	32 767	16
i32	-2.1×10^9	2.1×10^9	32
i64	-9.2×10^{18}	9.2×10^{18}	64
i128	$\approx -1.7 \times 10^{38}$	$\approx 1.7 \times 10^{38}$	128
isize	platform	platform	32/64



Mathematical constants - `std::f64::consts`:

π , e , $\sqrt{2}$, $\ln(10)$, etc.

```
use std::f64::consts;

fn main() {
    println!("π          = {:.10}", consts::PI);
    println!("τ = 2π      = {:.10}", consts::TAU);
    println!("e           = {:.10}", consts::E);
    println!("√2          = {:.10}", consts::SQRT_2);
    println!("ln 2        = {:.10}", consts::LN_2);
    println!("ln 10       = {:.10}", consts::LN_10);
    println!("1/π         = {:.10}", consts::FRAC_1_PI);
    println!("1/√2        = {:.10}", consts::FRAC_1_SQRT_2);
    println!("log10(2)    = {:.10}", consts::LOG10_2);
}
```

```
π          = 3.1415926536
τ = 2π      = 6.2831853072
e           = 2.7182818285
√2          = 1.4142135624
ln 2        = 0.6931471806
ln 10       = 2.3025850930
1/π         = 0.3183098862
1/√2        = 0.7071067812
log10(2)    = 0.3010299957
```

```
use std::f64::consts;

fn main() {
    let radius = 4.234;
    let diameter = 2.0 * radius;
    let area = consts::PI * radius * radius; // πr²
    println!("r={radius} d={diameter} area={area:.3}");
}
```



```
use std::f64::consts;
fn main() {
    let inf      = f64::INFINITY;
    let neg_inf  = f64::NEG_INFINITY;
    let nan      = f64::NaN;

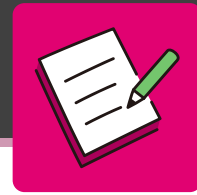
    println!(" ∞  = {inf}");           // inf
    println!("-∞  = {neg_inf}");       // -inf
    println!("NaN = {nan}");           // NaN

    // Arithmetic with special values
    println!("{}", 1.0_f64 / 0.0);     // inf
    println!("{}", -1.0_f64 / 0.0);   // -inf
    println!("{}", 0.0_f64 / 0.0);    // NaN

    // NaN is never equal - even to itself!
    println!("{}", f64::is_nan(f64::NaN)); // true
    println!("{}", f64::NaN.is_nan());      // true
    println!("{}", inf.is_infinite());      // true
    println!("{}", 42.0_f64.is_finite());   // true
}
```

∞ , $-\infty$, NaN

```
∞    = inf
-∞   = -inf
NaN  = NaN
inf
-inf
NaN
true
true
true
true
```



Using **only** `const` declarations (no `let` bindings), build a tiny time budget:

1. Declare the base constants `SECS_PER_MINUTE`, `MINS_PER_HOUR`, `HOURS_PER_DAY` and `DAYS_PER_YEAR`.
2. Derive `SECS_PER_YEAR` as a **computed constant expression** from the constants above.
3. In `main`, print `SECS_PER_YEAR` and report whether it fits in a `u16` and in a `u32`, using the associated boundary constants `u16::MAX` and `u32::MAX`.

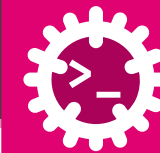


Use `SCREAMING_SNAKE_CASE` names and pick an integer type wide enough to hold the result.



Notations (Literals)

Expressing values directly in source code



Decimal, Hex, Octal, Binary, Byte

Supports five **numeric bases** for integer literals

```
fn main() {
    let decimal = 98_222;    // base 10
    let hex     = 0xFF;      // base 16, prefix 0x
    let octal   = 0o77;      // base 8,  prefix 0o
    let binary  = 0b1111_0000; // base 2,  prefix 0b
    let byte: u8 = b'A';     // ASCII byte literal

    println!("decimal : {decimal}"); // 98222
    println!("hex      : {hex}");    // 255
    println!("octal    : {octal}");  // 63
    println!("binary   : {binary}"); // 240
    println!("byte     : {byte}");   // 65

    // All represent the same value
    assert_eq!(0xFF_u8, 0b1111_1111_u8);
    assert_eq!(0o377_u16, 255_u16);
}
```

Number separator _ improves readability:

```
let million: u64    = 1_000_000;
let pi_approx: f64  = 3.141_592_653;
let flags: u8       = 0b0000_1111;
let ip_mask: u32    = 0xFF_FF_00_00;
```

Type suffixes make the type explicit:

```
let a: u8  = 57;    // unsigned 8-bit
let b: i32 = -100;   // signed 32-bit
let c: f32 = 1.5;    // 32-bit float
let d: i64 = 1_000;  // large signed integer
let e      = 57_i16; // explicit i16 suffix
```

Default types when no suffix is given: integer i32,
float f64, byte u8



Variables

Bindings: immutable by default



In Rust, bindings are **immutable by default**. The compiler **enforces** this at compile time:

```
fn main() {  
    let x = 5;  
    println!("x = {x}");    // x = 5  
    // This does NOT compile:  
    x = 6;  
}
```

Why immutable by default?

- Prevents accidental mutation bugs
- Enables safe concurrent access
- Communicates intent explicitly
- Allows aggressive compiler optimizations
- Forces deliberate design choices

```
error[E0384]: cannot assign twice to  
             immutable variable `x`  
  --> src/main.rs:4:5  
2 |         let x = 5;  
  |         -  
  |         |  
  |         first assignment to `x`  
  |         help: consider making this binding mutable: `mut x`  
4 |         x = 6;  
  |         ^^^^^ cannot assign twice to immutable variable
```



The compiler is your **friend** - it gives clear error messages and suggestions to fix issues.



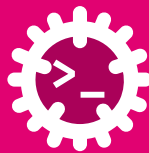
Add mut to allow reassignment. But **start immutable, add mut only when needed.**

```
struct Sensor { value: f64, unit: &'static str }
fn main() {
    // Mutable variable - can be reassigned
    let mut count = 0_u32;
    count += 1;
    println!("count = {count}"); // count = 1
    // Mutable bindings can also hold complex types
    let mut name = String::from("Alice");
    name.push_str(" Smith");
    println!("{name}");           // Alice Smith
    // Mutable structs: `mut` applies to all fields
    let mut s = Sensor { value: 20.0, unit: "°C" };
    // Simulate readings
    s.value += 3.5;
    s.value -= 1.2;
    println!("{:.1} {}", s.value, s.unit); // 22.3 °C
}
```

Real-world example - accumulator loop:

```
fn main() {
    let readings = [17.3, 19.1];
    let mut sum = 0.0_f64;
    for &r in &readings {
        sum += r;
        println!("reading: {r:.1}, sum: {sum:.1}");
    }
    let avg = sum / readings.len() as f64;
    println!("average: {avg:.1}");
}
```

```
reading: 17.3, sum: 17.3
reading: 19.1, sum: 36.4
average: 18.2
```



Rust infers types from **context** - no annotation needed when unambiguous:

Basic inference from value:

```
fn main() {  
    let a = 42;           // i32   (integer default)  
    let b = 3.14;         // f64   (float default)  
    let c = true;         // bool  
    let d = 'z';          // char  
    let e = "hello";      // &str  
    let f = 42_u8;        // u8  
    let g = 42_i128;      // i128  
    // Inferred from another binding  
    let x = 100;  
    let y = x;            // i32 - same as x  
    // Inferred from arithmetic context  
    let sum = a + y;      // i32  
    println!("{a} {b} {c} {d} {e} {f} {g} {sum}");  
}
```

Same code with explicit type annotations:

```
fn main() {  
    let a: i32 = 42;  
    let b: f64 = 3.14;  
    let c: bool = true;  
    let d: char = 'z';  
    let e: &str = "hello";  
    let f: u8 = 42;  
    let g: i128 = 42;  
    let x: i32 = 100;  
    let y: i32 = x;  
    let sum: i32 = a + y;  
    println!("{a} {b} {c} {d} {e} {f} {g} {sum}");  
}
```



Shadowing lets you reuse a name with a **new binding** - even with a different type:

```
fn main() {  
    let x = 5;           // x: i32 = 5  
    let x = x + 1;       // x: i32 = 6 (new binding)  
    {  
        let x = x * 2;   // x: i32 = 12 (inner scope)  
        println!("inner x = {x}"); // 12  
    }  
    println!("outer x = {x}"); // 6 (outer still 6)  
    // Shadow with a completely different TYPE  
    let spaces = " ";    // &str  
    let spaces = spaces.len(); // usize = 3  
    println!("spaces = {spaces}");  
}
```

Shadowing vs mut:

```
// With mut: same type, reassignable  
let mut y = 5;  
y = y + 1;    // ok - same type i32  
// y = "six"; // ERROR: type mismatch  
  
// With shadowing: can change type!  
let z = 5;  
let z = z + 1; // new binding, still i32  
let z = "six"; // new binding, now &str  
println!("{z}");
```



What prints the program out?



Two ways to say “I don’t need this” - keep the value but silence the warning, or throw it away:

```
fn main() {  
    // _name: BOUND but unused → no warning  
    let _guard = setup(); // kept alive, name silenced  
    let count = 5;        // ⚠ warning: unused  
    `count`  
    let _count = 5;        // no warning  
    // bare _ : NOT bound → value thrown away  
    let _ = returns_result(); // ignore the Result  
    let (x, _) = (1, 2);      // keep x=1, drop the 2  
    for _ in 0..3 {           // loop 3x  
        println!("tick");    // index not used  
    }  
}
```

`_name` **vs** `_` - the key difference:

- `_name` - a **real binding**; the value **lives** to end of scope, the leading `_` only mutes the “unused” warning
- `_` - **not a binding**; the value is **dropped immediately**, nothing is named
- So `let _g = guard();` **holds** the guard; `let _ = guard();` drops it **now**
- `_` also works in patterns: `match`, `if let`, tuples, closures - “match anything, bind nothing”



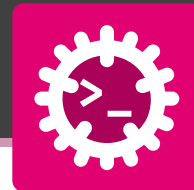
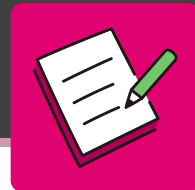
When does the Guard get dropped: with `let _g =` versus `let _ =`?



Lifetime is a compile-time concept that tracks **how long references are valid**. It ensures memory safety without a garbage collector.

In the example below, the inner `x` shadows the outer `x`, but they have different lifetimes - the inner `x` only exists within the inner block and is dropped when the block ends. The outer `x` remains unaffected and valid throughout the entire `main` function.

```
fn main() {  
    let x = 5;           // -----+--- b'  
    let x = x + 1;       //      |  
    {                   //      |  
        let x = x * 2;   // --+--- a'  
        println!("inner x = {x}"); //      |  
    }                   // --+--- a' ends, inner x dropped |  
    println!("outer x = {x}"); //      |  
}                       // -----+--- b' ends, outer x dropped
```



Write a small `main` that exercises bindings and shadowing:

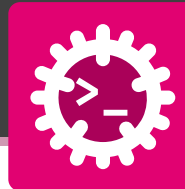
1. Declare an immutable `let level = 1`, then explain in one comment why `level = 2`; would **not compile**
2. Print value of variable `level` with `println!("{level}")`
3. Declare a mutable `let mut score = 0`, then add 10 to it **twice**
4. Remove the warning: variable `score` is assigned to, but never used
5. Starting from the string `" 42 "`, use **shadowing** to reuse the same name: first trim it with `.trim()`, then parse it into an `i32` with `.parse().unwrap()`
6. Print the value of the `i32` variable

Lean on **type inference** and only annotate a type where the compiler needs it.



Primitive Types

Scalar and compound built-in types



Type	Min	Max	Typical use
u8	0	255	bytes, RGB, ASCII
u16	0	65 535	network ports
u32	0	4.3×10^9	file sizes, colours
u64	0	1.8×10^{19}	large counters, hashes
u128	0	$\approx 3.4e38$	UUIDs, checksums
usize	0	platform	lengths, indices
i8	-128	127	small deltas, sensor raw
i16	-32 768	32 767	audio samples
i32	-2.1×10^9	2.1×10^9	default integer
i64	-9.2×10^{18}	9.2×10^{18}	timestamps, large IDs
i128	$\approx -1.7e38$	$\approx 1.7e38$	cryptography, finance
isize	platform	platform	signed index offsets

`isize` / `usize` match the **pointer width** and are platform dependent (32 or 64 bit):

```
fn main() {  
    let v = vec![10, 20, 30, 40];  
    // len() returns usize  
    let len: usize = v.len();  
    println!("last: {}", v[len - 1]); // 40  
    // Signed arithmetic on indices  
    let offset: isize = -2;  
    let i = (len as isize + offset) as usize;  
    println!("v[{}] = {}", i, v[i]); // v[2] = 30  
    // Platform size  
    println!("usize = {} bits", usize::BITS);  
    // 64 on modern machines  
}
```

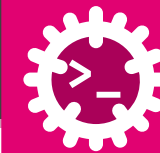


Type	Width	Precision	Range (abs)
f32	32 bit	7 decimal dig	1.2e-38 to 3.4e38
f64	64 bit	15 decimal dig	2.2e-308 to 1.8e308

```
fn main() {  
    // f64 is the default float type  
    let y: f64 = 3.14159265358979;  
  
    let area = std::f64::consts::PI * 5.0_f64.powi(2);  
    println!("area: {area:.4}");    // area: 78.5398  
  
    // Precision difference  
    let a: f32 = 0.1 + 0.2;  
    let b: f64 = 0.1 + 0.2;  
    println!("f32: {a:.20}"); // 0.30000001192092895508  
    println!("f64: {b:.20}"); // 0.30000000000000004441  
}
```

Common math methods:

```
let x = 2.0_f64;  
  
println!("{}", x.sqrt());    // 1.4142135623730951  
println!("{}", x.powi(10));  // 1024.0 (integer exp)  
println!("{}", x.powf(0.5)); // 1.4142135623730951  
println!("{}", x.ln());      // 0.6931471805599453  
println!("{}", x.log2());    // 1.0  
println!("{}", x.log10());   // 0.30102999566398114  
println!("{}", x.sin());     // 0.9092974268256817  
println!("{}", x.cos());     // -0.4161468365471424  
println!("{}", x.abs());     // 2.0  
println!("{}", (-3.7_f64).ceil()); // -3.0  
println!("{}", (-3.7_f64).floor()); // -4.0  
println!("{}", (-3.7_f64).round()); // -4.0  
println!("{}", (-3.7_f64).trunc()); // -3.0
```



bool - exactly 1 byte, values true or false:

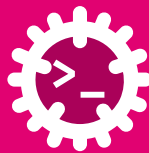
```
fn main() {  
    let t: bool = true;  
    let f      = false; // type inferred  
  
    // Produced by comparison operators  
    let is_adult = 18 ≥ 18;           // true  
    let is_zero  = 42 == 0;          // false  
    let in_range = 5 < 10 && 10 < 20; // true  
  
    // Used in control flow  
    if is_adult {  
        println!("welcome");  
    }  
  
    // Casting to integer  
    println!("{}", true as u8);      // 1  
    println!("{}", false as u8);     // 0  
}
```

Unit type () - empty tuple, zero bytes, “no value”:

```
// Functions without → return ()  
fn greet(name: &str) {  
    println!("Hello, {name}!");  
}  
  
fn main() {  
    let result: () = greet("Alice");  
    println!("{:?}", result); // ()  
}
```



?: is the debug formatter, which prints values in a human-readable form. For `()`, it simply prints `()`.



Tuple - ordered, fixed-size group of **heterogeneous** values:

```
fn main() {
    // Type annotation: (T1, T2, T3, ...)
    let point: (i32, i32)      = (10, 20);
    let color: (u8, u8, u8)    = (255, 128, 0); // RGB
    let mixed: (i32, f64, bool) = (42, 3.14, true);

    // Access by zero-based index (.0, .1, ...)
    println!("{}", point.0);    // 10
    println!("{}", color.2);    // 0
    println!("{}", mixed.1);    // 3.14

    // Tuples are Copy if all fields are Copy
    let a = point;
    let b = point; // both valid
    println!("{}", a.0, b.0); // 10 10
}
```

Destructuring - unpack a tuple into named variables:

```
fn main() {
    let point = (10, 20);
    let color = (255_u8, 128_u8, 0_u8);
    let mixed = (42, 3.14, true);

    // bind all fields
    let (x, y)      = point;
    let (r, g, b)   = color;

    // _ discards a field
    let (n, _, ok)  = mixed;
    println!("{x},{y} r={r} ok={ok} n={n}");

    // Nested tuples
    let nested = ((1, 2), (3, 4));
    println!("{}", nested.0.1); // 2 (first tuple, second
                                // field)
}
```



Default behaviour depends on build mode. Use explicit methods for full control:

```
fn main() {
    let x: u8 = 255;

    let bad = x + 1; // debug build: panics on overflow
    // release build: wraps silently (255 + 1 = 0)

    // — Explicit overflow strategies —
    // wrapping_*: always wraps (modular arithmetic)
    println!("{}", x.wrapping_add(1));    // 0
    println!("{}", x.wrapping_add(10));   // 9

    // saturating_*: clamps to boundary
    println!("{}", x.saturating_add(1));   // 255
    println!("{}", 0_u8.saturating_sub(5)); // 0

    // checked_*: returns Option
    println!("{}", x.checked_add(1));      // None
    println!("{}", x.checked_add(0));      // Some(255)
}
```

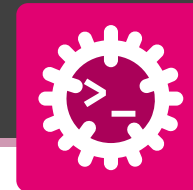
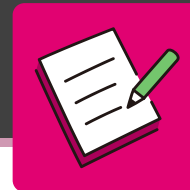
When does overflow matter?

- Embedded systems with small integer buffers
- Network protocol byte counters
- Cryptographic arithmetic
- Loop index arithmetic

Summary of overflow methods:

Method	Behaviour
<code>wrapping_add</code>	wrap around (modular)
<code>saturating_add</code>	clamp to MIN/MAX
<code>checked_add</code>	return <code>Option<T></code>
<code>overflowing_add</code>	return <code>(T, bool)</code>

All four families exist: `_add`, `_sub`, `_mul`, `_div`, `_shl`, `_shr`, `_neg`.



Read the code below (don't compile it yet) and spot the **type errors**. For each line, say **why** the compiler rejects it and how you would fix it.

```
fn main() {  
    let a: u8 = 300;  
    let b: u8 = 10;  
    let sum: u16 = a + b;  
    let ratio = 7 / 2.0;  
    let flag: bool = 1;  
    let point = (1, 2.0);  
    let z: i32 = point.1;  
}
```



Operators

Arithmetic, Comparison, Logical, and Bitwise



Arithmetic - +, -, *, /, %:

```
fn main() {
    let a: i32 = 17;
    let b: i32 = 5;

    println!("{a} + {b} = {}", a + b); // 22
    println!("{a} - {b} = {}", a - b); // 12
    println!("{a} * {b} = {}", a * b); // 85
    println!("{a} / {b} = {}", a / b); // 3 ← int
    println!("{a} % {b} = {}", a % b); // 2

    // Integer division truncates toward zero
    println!("{}", -7_i32 / 2); // -3 (not -4!)
    println!("{}", -7_i32 % 2); // -1

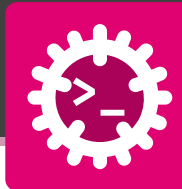
    // Compound assignment
    let mut n = 10;
    n += 5; n -= 3; n *= 2; n /= 4; n %= 4;
    println!("{n}"); // 2
}
```

Comparison - return bool:

```
fn main() {
    let x = 42;
    println!("{}", x == 42); // true (equal)
    println!("{}", x != 42); // false (not equal)
    println!("{}", x > 10); // true (greater than)
    println!("{}", x < 100); // true (less than)
    println!("{}", x >= 42); // true (greater or
    equal)
    println!("{}", x <= 42); // true (less or equal)
}
```

Operator precedence:

Highest $\Rightarrow$ Lowest (*, /, %),
(+, -), (<<, >>), (&), (^), (|), (==, !=, <, >, ≤, ≥), (&&), (||)



Logical Operators - work on `bool` values.

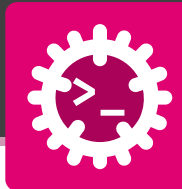
`&&` AND, `||` OR, `^` (XOR) and `!` NOT:

```
fn main() {
    let a: bool = true;
    let b: bool = false;

    println!("{}", a && b); // false (AND)
    println!("{}", a || b); // true  (OR)
    println!("{}", a ^ b);  // true  (XOR)
    println!("{}", !a);     // false (NOT)
}
```

```
fn main() {
    // && eval from left to right
    // RHS not evaluated if result is known
    let x: i32 = 0;
    // Safe: 10/x never runs when x == 0
    if x != 0 && 10 / x > 1 {
        println!("big quotient");
    } else {
        println!("x is zero, skipped division");
    }

    // || RHS not evaluated if LHS is true
    fn expensive() → bool { println!("called!"); true }
    if true || expensive() {
        println!("short-circuited");
    }
    // "short-circuited" - expensive() never ran
}
```



Bitwise operators - work on integer bits:

```
fn main() {
    let a: u8 = 0b1100_1010; // 202
    let b: u8 = 0b1010_1100; // 172

    println!("{:08b}", a & b); // 10001000 AND
    println!("{:08b}", a | b); // 11101110 OR
    println!("{:08b}", a ^ b); // 01100110 XOR (toggle)
    println!("{:08b}", !a);    // 00110101 NOT (bitflip)

    // Shift operators
    let flags: u8 = 0b0000_0001;
    println!("{:08b}", flags << 3); // 00001000 (×8)
    println!("{:08b}", flags >> 1); // 00000000 (÷2)
}
```

Practical: hardware register flags

```
fn main() {
    const LED_ON:    u8 = 0b0000_0001;
    const MOTOR_ON:  u8 = 0b0000_0010;
    const ALARM_ON:  u8 = 0b0000_0100;

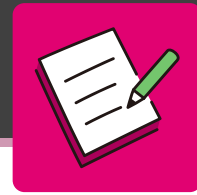
    let mut reg: u8 = 0b0000_0000;

    // set bits → 0b0000_0011
    reg |= LED_ON | MOTOR_ON;

    // flip bits → 0b0000_0111
    reg ^= ALARM_ON;

    // clear bits → 0b0000_0011
    reg &= !ALARM_ON;

    // 0b00000011 (LED & MOTOR)
    println!("0b{:08b}", reg);
}
```



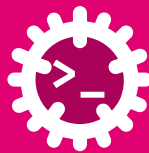
Trace the code by hand and write down the **output value** printed by each line. Mind **integer division**, **bitwise** operators and **operator precedence**.

```
fn main() {  
    let a = 17;  
    let b = 5;  
    println!("{}", a / b);           // 1  
    println!("{}", a % b);           // 2  
    println!("{}", a / b * b);       // 3  
  
    let x = 0b1010;                  // 10  
    let y = 0b0110;                  // 6  
    println!("{}", x & y);           // 4  
    println!("{}", x | y);           // 5  
    println!("{}", x ^ y);           // 6  
    println!("{}", x << 1);          // 7  
  
    println!("{}", true && false || !false); // 8  
    println!("{}", 3 + 4 * 2 > 10);   // 9  
}
```



Type Conversion

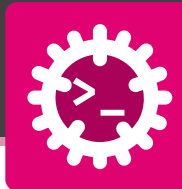
Explicit, safe, and idiomatic conversions



as performs **numeric casts** - fast but potentially **lossy**:

```
fn main() {  
    // Widening  
    // (always safe)  
    let small: u8 = 200;  
    let wide: u32 = small as u32;  
    println!("{wide}");           // 200  
  
    // Narrowing  
    // (truncates bits - potential loss!)  
    let big: u32 = 300;  
    let tiny: u8 = big as u8;      // 300 - 256 = 44  
    println!("{tiny}");           // 44  
  
    // Float → integer  
    // (truncates toward zero, no round)  
    let f: f64 = 3.99;  
    let i: i32 = f as i32;  
    println!("{i}");              // 3  
  
    // Negative float → unsigned: saturates
```

```
    let neg: f64 = -1.5;  
    let u: u32 = neg as u32;  
    println!("{u}");              // 0  
  
    // char ↔ u32  
    let c = 'A';  
    println!("{}", c as u32);     // 65  
    println!("{}", 66_u8 as char); // B  
}
```



Why is as dangerous?

```
// Example: reading sensor data
fn read_sensor() → u32 { 300 } // returns 300
fn process() {
    let raw: u32 = read_sensor();
    // Silent data loss - no warning!
    let byte: u8 = raw as u8; // 44, not 300!
    // Use in protocol field → corrupted packet
    println!("sending byte: {byte}");
}
```

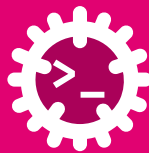
```
300 = 0b00000001_00101100
           ^^^^^^^
```

keep these 8 bits

```
0b00101100 = 44
```

Rule of thumb for as:

Situation	Recommendation
Widening (u8 → u32)	as is always safe
Float ↔ integer	as is OK (truncates)
Narrowing (u32 → u8)	prefer try_into()
Pointer casts	as - systems code only



Fallible - returns Result, never silently loses data:

```
use std::convert::TryFrom;
use std::convert::TryInto;

fn main() {
    // TryFrom: explicit, static method
    let big: i32 = 300;
    match u8::try_from(big) {
        Ok(v) => println!("ok: {v}"),
        Err(e) => println!("error: {e}"),
    } // error: out of range integral type conversion

    // TryInto: method syntax (often preferred)
    let ok: i32 = 200;
    let result: Result<u8, _> = ok.try_into();
    println!("{:?}", result); // Ok(200)

    let bad: i32 = -5;
    let result: Result<u8, _> = bad.try_into();
    println!("{:?}", result); // Err(TryFromIntError(()))
}
```

Infallible - From / Into (always succeeds):

```
// From: explicit construction
let s = String::from("hello"); // &str -> String
let n = i64::from(42_i32);      // i32 -> i64

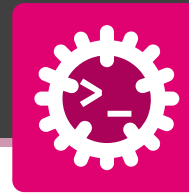
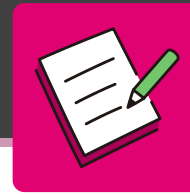
// Into: derived automatically from From
let s: String = "world".into(); // &str -> String
let n: i64 = 42_i32.into();     // i32 -> i64

// Custom conversion
struct Celsius(f64);
struct Kelvin(f64);

impl From<Celsius> for Kelvin {
    fn from(c: Celsius) -> Self {
        Kelvin(c.0 + 273.15)
    }
}

let boiling = Kelvin::from(Celsius(100.0));
println!("{:.2} K", boiling.0); // 373.15 K

// Into is auto-derived
let freezing: Kelvin = Celsius(0.0).into();
println!("{:.2} K", freezing.0); // 273.15 K
```



Trace each as cast and write down the **output value**. Watch for **narrowing**, **truncation toward zero** and **saturation**.

```
fn main() {  
    println!("{}", 300_i32 as u8); // 1  
    println!("{}", 3.99_f64 as i32); // 2  
    println!("{}", -1.5_f64 as u32); // 3  
    println!("{}", 'A' as u32); // 4  
    println!("{}", 66_u8 as char); // 5  
    println!("{}", 255_u8 as i8); // 6  
    println!("{}", -1_i8 as u8); // 7  
}
```

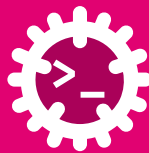


as never fails - it silently reinterprets or clamps. Prefer TryFrom/TryInto when loss must be caught.



Text Types

char, &str, String



char - a single **Unicode Scalar Value (USV)**, always 4 bytes wide:

```
fn main() {
    let letter: char = 'A';
    let emoji: char = '🦀';           // Ferris the crab!
    let accent: char = 'é';           // accented letter
    let japanese: char = '日';        // "day" in Japanese
    let ascii_a: i32 = 'A' as i32;    // 65 - see ASCII

    table

    // Escape Characters
    let newline: char = '\n';
    let carriage_return: char = '\r';
    let tab: char = '\t';
    let quote: char = '"';
    let backslash: char = '\\';
    let 7bit-unicode: char = '\u{7F}'; // DEL
    let 8bit-unicode: char = '\u{1F600}'; // 😄
    let 24bit-unicode: char = '\u{10FFFF}';
```

```
// Classification
println!("{}", 'a'.is_alphabetic()); // true
println!("{}", '5'.is_ascii_digit()); // true
println!("{}", '5'.is_alphanumeric()); // true
println!("{}", '2'.is_numeric()); // true
println!("{}", ' '.is_whitespace()); // true
println!("{}", 'A'.is_uppercase()); // true
println!("{}", 'z'.is_lowercase()); // true
println!("{}", 'A'.is_ascii()); // true

// Case conversion
println!("{}", 'a'.to_uppercase()); // A (it)
println!("{}", 'Z'.to_lowercase()); // z (it)

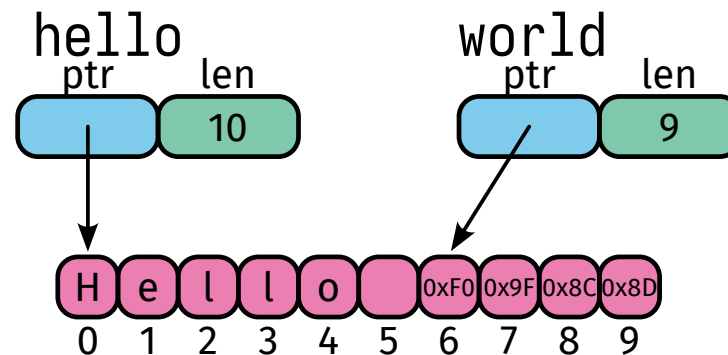
// Digit conversion
println!("{}", 'F'.to_digit(16)); // Some(15)
println!("{}", char::from_digit(10, 16)); // Some('a')
}
```



&str - an immutable **borrowed** view into string bytes.

```
fn main() {  
    // String literals have static lifetime  
    let hello: &str = "Hello 🌍";  
    // Byte length (not char count!)  
    println!("{}", hello.len());           // 10 (🌍 = 4 bytes)  
    // Char count  
    println!("{}", hello.chars().count()); // 7  
    // Slice (byte-indexed - must be on char boundaries!)  
    let world: &str = &hello[6..10];       // 6-9 (10 excluded)  
    println!("{}", world);                 // 🌍  
    // Iterate characters  
    for c in hello.chars().take(5) {  
        print!("{}", c); // H e l l o  
    }  
}
```

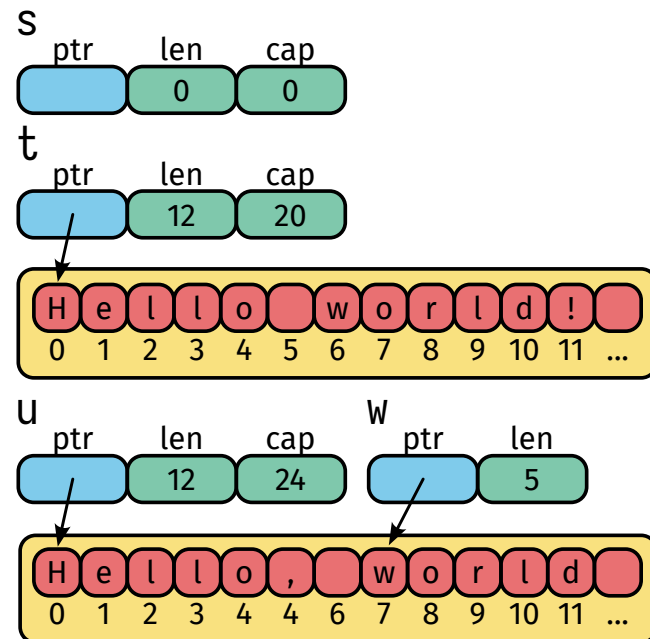
They are stored as literal data in the binary and have a 'static lifetime.





String - Heap-allocated, growable, UTF-8 text:

```
fn main() {  
    let s = String::new();           // empty string  
    // Append  
    let mut t = String::from("Hello"); // from literal  
    t.push(' ');                     // single char  
    t.push_str("world!");           // string slice  
    println!("{t}");                 // Hello world!  
    println!("len={} cap={}", t.len(), t.capacity());  
    // Insert  
    let mut u = String::from("Hello world!");  
    u.insert(5, ',');                // insert char at index  
    println!("{u}");                 // Hello, world!  
    // Remove  
    u.pop();                         // removes last char  
    println!("{u}");                 // Hello, world  
    // Slice  
    let w = &u[7..12];              // "world" (7-11)  
}
```

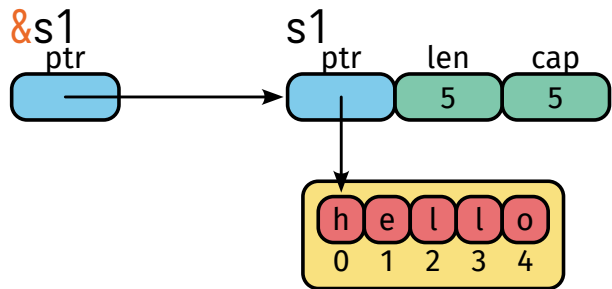


- `cap` is the max number of bytes the string can hold without reallocating
- `len` is the current number of bytes used

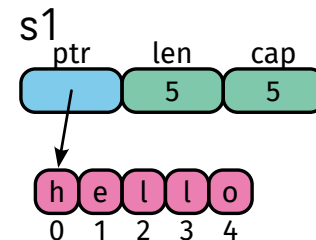


String	&str
Owned (on heap)	Borrowed (a view)
Mutable	Immutable
Has len, capacity	Has len only
Growable	Fixed slice
Allocates memory	Zero allocation

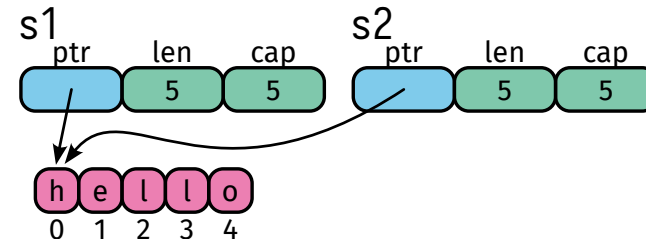
```
let s1 = String::from("hello");
&s1
```



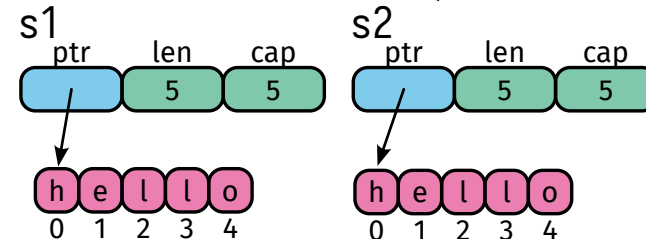
```
let s1 = "hello";
```

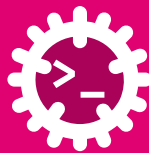


```
let s2 = s1;
```



```
let s2 = s1.clone();
```





Searching:

```
let s = "Hello, Rust world!";
println!("{}", s.contains("Rust")); // true
println!("{}", s.starts_with("Hello")); // true
println!("{}", s.ends_with("!")); // true
println!("{}", s.find("Rust")); // Some(7)
println!("{}", s.rfind('o')); // Some(13)
```

Replacing and splitting:

```
let r = s.replace("Rust", "Go");
println!("{}", r); // Hello, Go world!
let words: Vec<&str> = s.split_whitespace().collect();
println!("{}", words);
// ["Hello,", "Rust", "world!"]
let parts: Vec<&str> = "a,b,c".split(',').collect();
println!("{}", parts); // ["a", "b", "c"]
```

Trimming and transforming:

```
let padded = " hello world ";
println!("{}", padded.trim()); // 'hello world'
println!("{}", padded.trim_start()); // 'hello world '
println!("{}", padded.trim_end()); // ' hello world'
println!("{}", "hello".to_uppercase()); // HELLO
println!("{}", "WORLD".to_lowercase()); // world
println!("{}", "=".repeat(20)); // =====
```

Collecting:

```
// Chars to Vec
let chars: Vec<char> = "abc".chars().collect();
println!("{}", chars); // ['a', 'b', 'c']
// Join lines
let lines = ["line1", "line2", "line3"];
let joined = lines.join("\n");
println!("{}", joined); //line1\nline2\nline3
```



Value → String:

```
fn main() {  
    let n: i32 = 42;  
    let f: f64 = 3.14;  
    let b: bool = true;  
  
    // to_string() works on any Display type  
    let s1 = n.to_string(); // "42"  
    let s2 = f.to_string(); // "3.14"  
    let s3 = b.to_string(); // "true"  
    println!("{s1} {s2} {s3}");  
  
    // format! - more control  
    let msg = format!("n={n:05}, f={f:.2}");  
    println!("{msg}"); // n=00042, f=3.14  
}
```

String → value - parse():

```
let text = "42";  
let n: i32 = text.parse().expect("bad input");
```

&str → String - to_owned():

```
let a = "hi".to_owned(); // clearest for &str  
let b = "hi".to_string(); // more general  
let c = String::from("hi"); // explicit  
let d: String = "hi".into(); // inferred  
println!("{a} {b} {c} {d}"); // hi hi hi hi
```

When to use which:

Method	Best for
to_string()	any Display type → String
to_owned()	&str specifically → String
String::from	explicit, readable constructor
.into()	idiomatic generic / trait code
.parse()	text → numeric or other types
format!(...)	multi-value or formatted strings



Arguments

```
let s = format!("Hello, {}, the answer is {}", "Alice", 42);           // Positional Arguments
let s = format!("Hello, {name}, the answer is {age}", name = "Bob", age = 7); // Named Arguments
let s = format!("{0} is {1} years old. {0} loves Rust!", "Carol", 25); // Reusing Arguments
```

Align

```
format!("{:<8}", 1); // 1..... - left aligned
format!("{:>8}", 1); // .....1 - right aligned
format!("{:^8}", 1); // ...1.... - center aligned
```

Padding

```
format!("{:*<8}", 1); // 1***** - left align, padding "*"
format!("{:0>8}", 1); // 00000001 - right align, padding "0"
format!("{:>08}", 1); // 00000001 - right align, padding leading "0"
```



```
format!("{: >8}", 1); //      1 - right align, padding " "  
format!("{: #^8}", 1); // ###1#### - right align, padding "#"
```

Sign

```
format!("{: >+8}", 1); //      +1 - right align, sign +  
format!("{: >8}", -1); //      -1 - right align, sign -  
format!("{: >+ #08}", 1); // +0000001 - right align, sign +, padding leading "0"  
format!("{: 0 >+ #8}", 1); // 000000+1 - right align, padding "0", sign +
```

Precision

```
format!("{: * <8}", 1); // 1***** - left align, padding "*"   
format!("{: 0 >8}", 1); // 00000001 - right align, padding "0"   
format!("{: >08}", 1); // 00000001 - right align, padding leading "0"   
format!("{: >8}", 1); //      1 - right align, padding " "   
format!("{: #^8}", 1); // ###1#### - right align, padding "#"
```



Type

```
format!("{:}", E as u32); // 2 - Integer (follows type)
format!("{:}", E); // 2.718281828459045 - Float (follows type)
format!("{:b}", 49374); // 1100000011011110 - Binary
format!("{:#b}", 49374); // 0b1100000011011110 - Binary alternative (adds "0b")
format!("{:#034b}", 49374); // 0b00000000000000001100000011011110 - Binary 32bit (adds "0b" and 32 digits)
format!("{:o}", 49374); // 140336 - Octal
format!("{:#o}", 49374); // 0o140336 - Octal alternative (adds "0o")
format!("{:#018o}", 49374); // 0o00000000000140336 - Octal 16 digits (adds "0o" and 16 digits)
format!("{:x}", 49374); // c0de - Hex lower case
format!("{:#x}", 49374); // 0xc0de - Hex lower case alternative (adds "0x")
format!("{:#010x}", 49374); // 0x0000c0de - Hex 8 digits (adds "0x" and 8 digits)
format!("{:X}", 49374); // C0DE - Hex upper case
format!("{:#X}", 49374); // 0xC0DE - Hex upper case alternative (adds "0X")
format!("{:#010X}", 49374); // 0x0000C0DE - Hex 8 digits (adds "0X" and 8 digits)
format!("{:e}", E.pow(123.0)); // 2.6195173187490456e53 - Scientific notation lower case (adds "e")
format!("{:E}", E.pow(123.0)); // 2.6195173187490456E53 - Scientific notation upper case (adds "E")
format!("{:p}", &1); // 0x1021872c0 - Pointer address
```


Rust

```
formatting = [argument][:][[fill]align][sign][#][0][width][.precision][type]
argument   = integer | identifier
fill       = <any character>
align      = "<" | ">" | "^"
sign       = "+"
"#"        = alternate formatting
"0"        = zero-padding
width      = integer
precision  = integer
type       = "b" | "o" | "x" | "X" | "e" | "E" | "p" | "?"
```

```
"{" [argument][":"][[fill]align[sign]"#"][width][precision][type]"}"
```

- padding "0"

- alternate formatting
- (human-readable)

- "" - default
- + - + prefix by tschinz

nbr of descendants - integer -

- < - left (default)
- > - right
- ^ - center

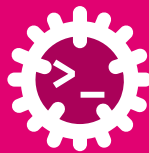
- any character for padding

- "" - follows arg sequence (default)
- integer - selects arg by position
- identifier - selects arg by name

"" - default follows type -
"b" - binary _____
"o" - octal _____
"x" - hex _____
"X" - Hex _____
"e" - scientific _____
"E" - Scientific _____
"p" - pointer addr _____
"? " - debug (trait req) ____

```
print!("{0:*>+10.3e}", std::f64::consts::E)
```

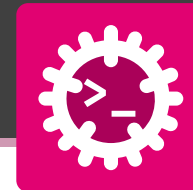
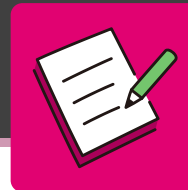
<https://wiki.zahno.dev/blog/2024/11/20/rust-string-formatting---an-ascii-art-cheatsheet.html>



Building a formatted table:

```
fn main() {  
    let items = [  
        ("Alice", 92, 98.7),  
        ("Bob", 87, 91.2),  
        ("Carol", 95, 99.1),  
    ];  
    // Header (left-align name, right-align numbers)  
    println!("| {:<5} | {:>5} | {:>6} |", "Name", "Score", "Pct");  
    // Alignment row (: marks alignment direction)  
    println!("|:{0:-<6}|{0:-<6}:|{0:-<7}:|", "");  
    // Data rows  
    for (name, score, pct) in items {  
        println!("| {:<5} | {:>5} | {:>5.1}% |", name, score, pct);  
    }  
}
```

	Name		Score		Pct	
	:		:		:	
	Alice		92		98.7%	
	Bob		87		91.2%	
	Carol		95		99.1%	



Part 1 - What is the output?

Trace the conversions and format specifiers.

```
fn main() {  
    let mut s = String::new();  
    s.push_str("Rust");  
    s.push('!');  
  
    let n: i32 = "42".parse().unwrap();  
    let label = "id".to_owned() + "-" + &n.to_string();  
  
    println!("{s} {}", s.len()); // a  
    println!("{label}");         // b  
    println!("{:>6}", "hi");     // c  
    println!("{:*<6}", "hi");    // d  
    println!("{:08.3}", 3.14159); // e  
    println!("{:#x}", 255);       // f  
}
```

Part 2 - Build the string

Given the bindings below, write a **single** format! call that produces **exactly** this line:

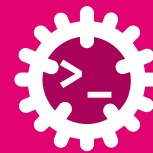
```
Bob      : 87
```

```
let name = "Bob"; // left-aligned, width 6  
let score = 87;   // right-aligned, width 3  
let line: String = format!("{: ? }");
```



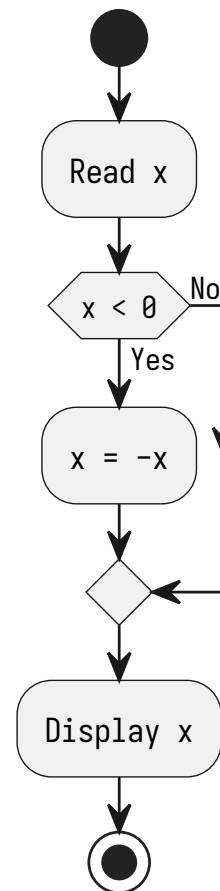
Control Flow

`if, else if, else, match, loopS, and more`



if is an **expression**:

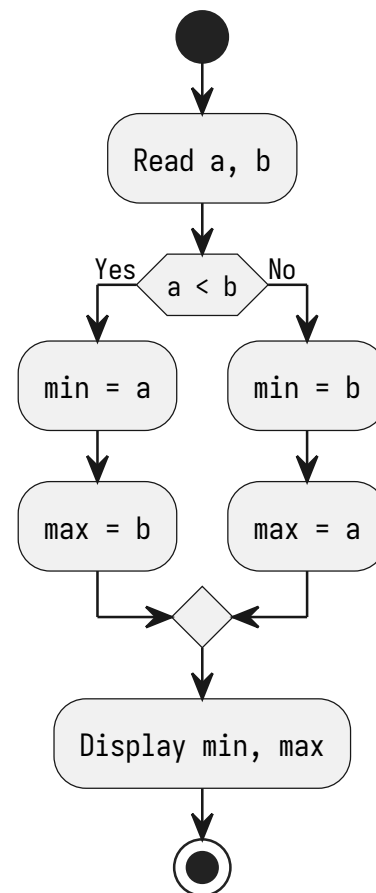
```
fn main() {  
    let mut x: i32 = -7;  
    // Statement form  
    if x < 0 {  
        x = -x; // make positive  
    }  
    println!("{x}"); // +7  
    // Expression form - both arms must return same type  
    x = if x < 0 { -x };  
    println!("{x}"); // +7  
}
```





if / else - conditional branch; if is an **expression**:

```
fn main() {  
    let (a, b): (i32, i32) = (42, 7);  
    let (max, min);  
    // Statement form  
    if a < b {  
        max = b; min = a;  
    } else {  
        max = a; min = b;  
    }  
    println!("Max {max}, Min {min}");  
    // Expression form - both arms must return same type  
    let (a, b): (i32, i32) = (7, 42);  
    let (max, min);  
    (max, min) = if a < b { (b, a) } else { (a, b) };  
    println!("Max {max}, Min {min}");  
}
```

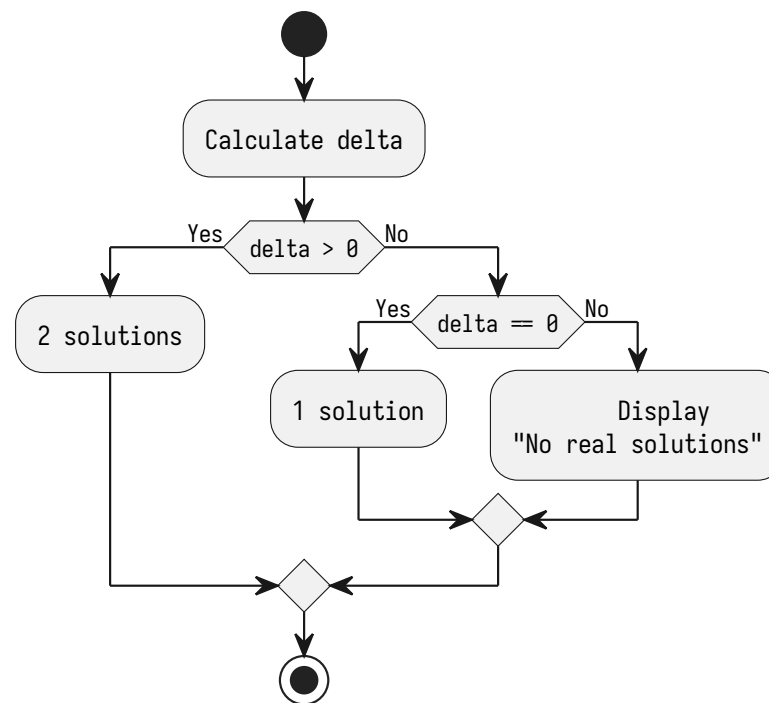


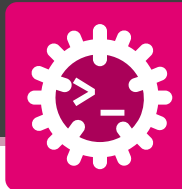


else if - evaluates branches top-to-bottom, first match wins:

```
fn main() {  
    // ax2 + bx + c = 0  
    let (a, b, c) = (1.0, -3.0, 2.0);  
    // Δ = b2 - 4ac  
    let delta = b * b - 4.0 * a * c;  
    if delta > 0.0 {  
        // x1,2 = (-b ± √Δ) / (2a)  
        let x1 = (-b + delta.sqrt()) / (2.0 * a);  
        let x2 = (-b - delta.sqrt()) / (2.0 * a);  
        println!("x1 = {x1}, x2 = {x2}");  
    } else if delta == 0.0 {  
        // x = -b / (2a)  
        let x = -b / (2.0 * a);  
        println!("x = {x}");  
    } else {  
        println!("No real solutions");  
    }  
}
```

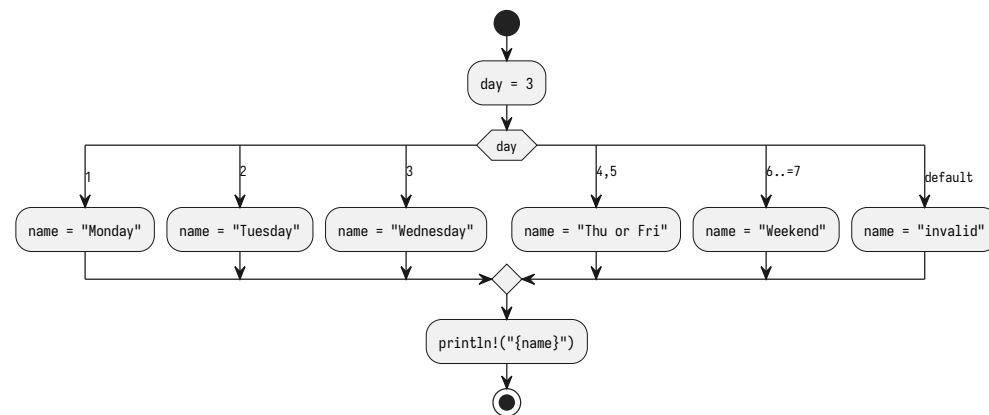
$$y = ax^2 + bx + c \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

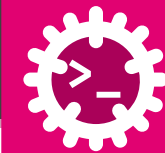




match - exhaustive multi-way branch on a value or pattern:

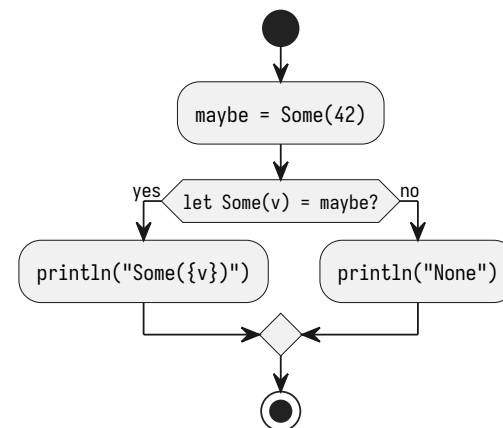
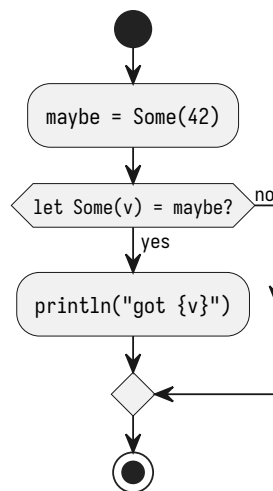
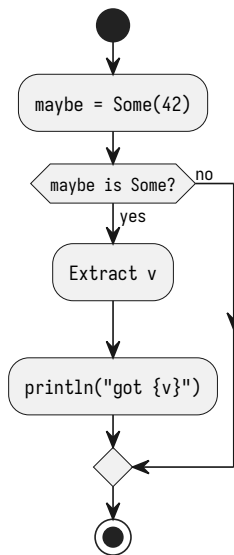
```
fn main() {  
    let day: u8 = 3;  
    let name = match day {  
        1      => "Monday",  
        2      => "Tuesday",  
        3      => "Wednesday",  
        4 | 5  => "Thu or Fri", // OR pattern  
        6..=7  => "Weekend",   // inclusive range  
        _      => "invalid",   // wildcard (must be last)  
    };  
    println!("{name}"); // Wednesday  
}
```





if let - concise pattern match for a **single** variant:

```
fn main() {  
    let maybe: Option<i32> = Some(42);  
    // Verbose: full match  
    match maybe {  
        Some(v) => println!("got {v}"),  
        None    => {},  
    }  
    // Concise: if let  
    if let Some(v) = maybe {  
        println!("got {v}"); // 42  
    }  
    // with else branch  
    if let Some(v) = maybe {  
        println!("Some({v})"); // Some(42)  
    } else {  
        println!("None");  
    }  
}
```

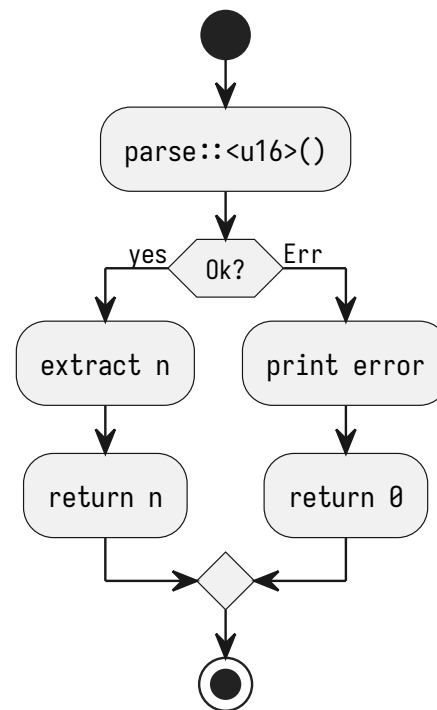


Don't worry it will get more clear with the introduction of enums later.



let else - bind a pattern or **diverge** (return / panic / break):

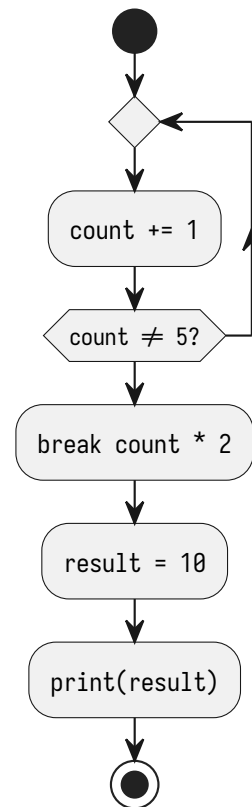
```
fn parse_port(s: &str) → u16 {  
    // Must diverge in the else branch  
    let Ok(n) = s.parse::<u16>() else {  
        eprintln!("invalid port: {s}");  
        return 0;           // diverge!  
    };  
    // n is in scope from here on  
    n  
}  
  
fn main() {  
    println!("{}", parse_port("8080")); // 8080  
    println!("{}", parse_port("abc"));  // 0  
}
```

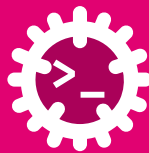




loop - runs forever; exit with **break**; can **return a value**:

```
fn main() {  
    let mut count = 0;  
    // loop is an expression - break carries the value  
    let result = loop {  
        count += 1;  
        if count == 5 {  
            break count * 2;    // returns 10  
        }  
    };  
    println!("{result}"); // 10  
    // Labelled loop  
    'retry: loop {  
        // ... break 'retry to exit by name  
        break 'retry;  
    }  
}
```





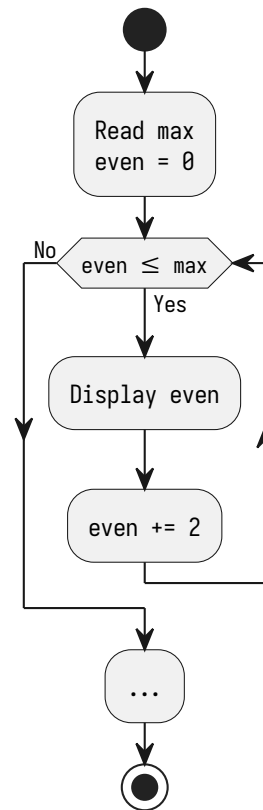
while - checks condition **before** each iteration:

Classic while

```
fn main() {  
    let max = 10;  
    let mut even = 0;  
    while even ≤ max {  
        println!("{even}");  
        even += 2;  
    }  
}
```

while let - loop while pattern matches

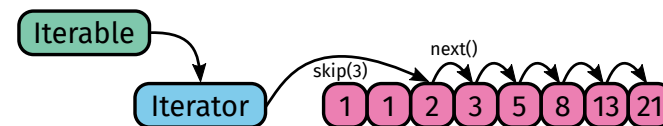
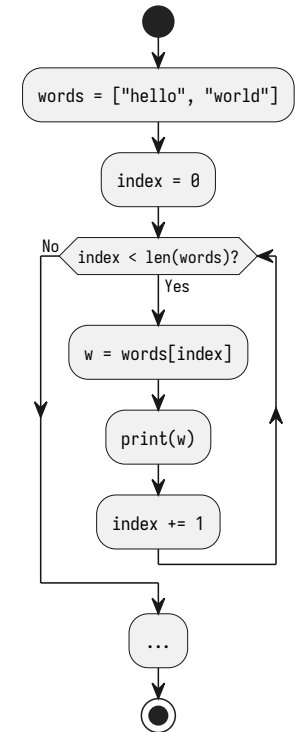
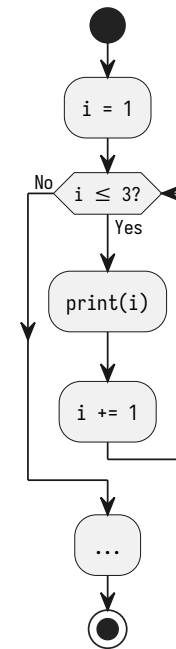
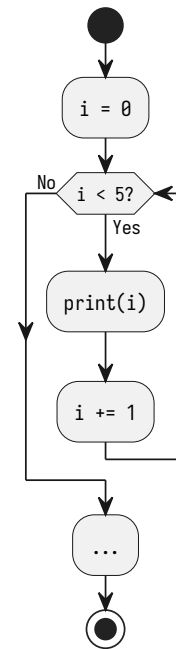
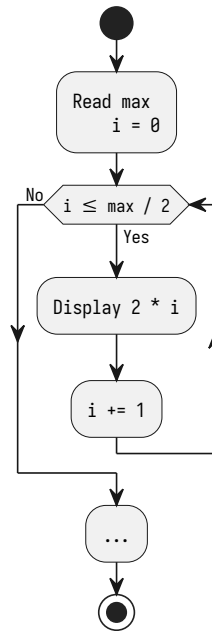
```
fn main() {  
    let mut stack = vec![3, 2, 1];  
    while let Some(top) = stack.pop() {  
        print!("{top} ");  
    }  
}
```





for - iterates over any IntoIterator

```
fn main() {  
    let max = 10;  
    for i in 0..=max / 2 {  
        println!("{}", 2 * i);  
    }  
    // Half-open range [0, 4]  
    for i in 0..5 {  
        print!("{}", i);    // 0 1 2 3 4  
    }  
    // Inclusive range [1, 3]  
    for i in 1..=3 {  
        print!("{}", i);    // 1 2 3  
    }  
    // Iterate over a collection  
    let words = ["hello", "world"];  
    for w in words {  
        println!("{}", w);  
    }  
}
```





```
fn main() {
    let words = ["alpha", "beta", "gamma"];
    let nums = [10, 20, 30];

    // enumerate() - (index, value)
    for (i, w) in words.iter().enumerate() {
        println!("{i}: {w}"); // 0: alpha ...
    }

    // zip() - pair two iterators
    for (w, n) in words.iter().zip(nums) {
        println!("{w}={n}"); // alpha=10 ...
    }

    // rev() - reverse a range
    for i in (1..=5).rev() {
        println!("{i} "); // 5 4 3 2 1
    }

    // step_by() - custom step (replaces i+=3)
    for i in (0..12).step_by(3) {
        println!("{i} "); // 0 3 6 9
    }
}
```

```
fn main() {
    let nums = [1, 2, 3, 4, 5, 6];

    // filter() - skip elements
    for x in nums.iter().filter(|&x| x % 2 == 0) {
        println!("{x} "); // 2 4 6
    }

    // map() - transform each element
    for x in nums.iter().map(|x| x * x) {
        println!("{x} "); // 1 4 9 16 25 36
    }

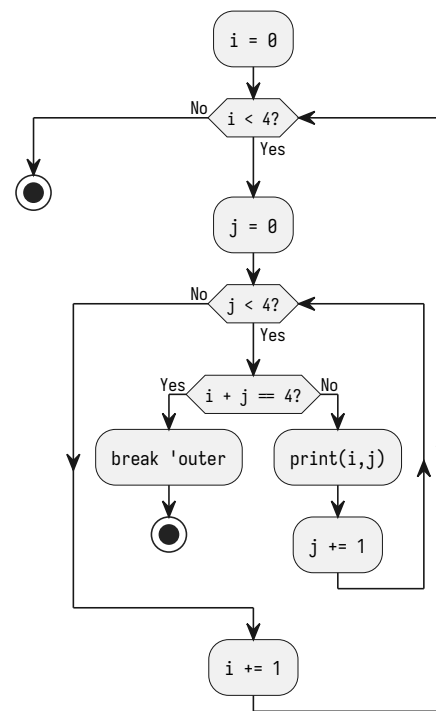
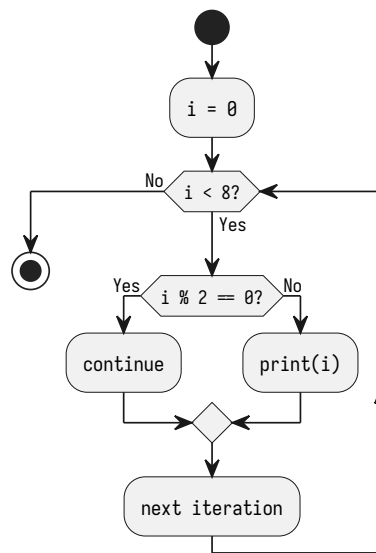
    // chain() - join two iterables
    let a = [1, 2];
    let b = [3, 4];
    for x in a.iter().chain(b.iter()) {
        println!("{x} "); // 1 2 3 4
    }

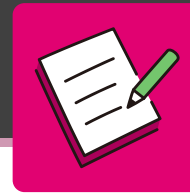
    // flatten() - one level of nesting
    let nested = [[1, 2], [3, 4]];
    for x in nested.iter().flatten() {
        println!("{x} "); // 1 2 3 4
    }
}
```



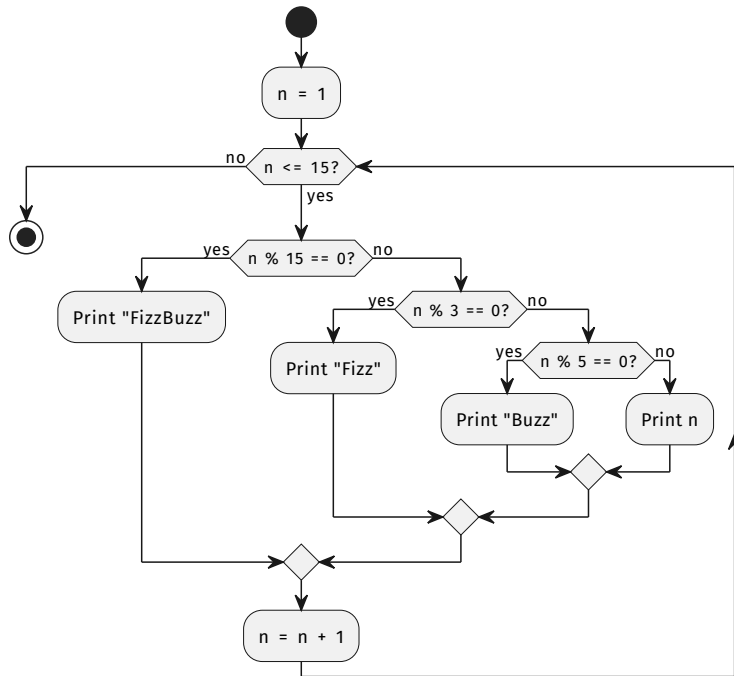
continue skips the rest of the body; **break** exits the loop:

```
fn main() {  
    // continue: skip even numbers  
    for i in 0..8 {  
        if i % 2 == 0 { continue; }  
        print!("{i} ");           // 1 3 5 7  
    }  
  
    // break with label: exit outer loop  
    'outer: for i in 0..4 {  
        for j in 0..4 {  
            if i + j == 4 { break 'outer; }  
            print!("{i},{j} ");  
        }  
    }  
}
```





Translate the flowchart below into Rust using a for loop and an if / else if / else chain.



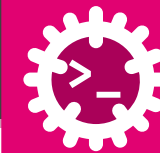
Order matters: the **most specific** test (% 15) must come **first**, otherwise % 3 would catch it.

Bonus: can you express the same logic with a single match?



Functions

Let's make spaghetti!



fn - explicit type on every parameter and on the return:

```
//      parameters (name: Type)
//      |          |          |  return type
fn add(a: i32, b: i32) → i32 {
    a + b    // implicit return: last expr, no `;`
}

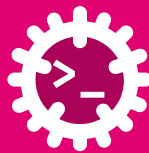
// no return type `→ T` means `→ ()` (unit)
fn greet(name: &str) {
    println!("Hello, {name}!");
}

// explicit early return
fn abs(x: i32) → i32 {
    if x < 0 { return -x; } // early exit
    x                      // implicit return
}

fn main() {
    println!("{}", add(3, 4)); // 7
    greet("Alice");           // Hello, Alice!
    println!("{}", abs(-9));  // 9
}
```

teaser: this is where functions can go

```
//      function name
//      |
//      'a      lifetime parameter
//      |
//      //      generic type T
//      //      first input slice
//      //      second input slice
//      //      return type:
//      //      a borrowed slice
fn longest<'a, T>(x: &'a [T], y: &'a [T]) → &'a [T]
where T: Ord,
//      T must be orderable, so slices [T] can be compared
//      trait bound
{
    if x > y { x } else { y }
}
```

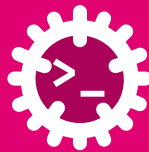


Single value and no value (()):

```
fn square(x: f64) → f64 { x * x }
// returns () implicitly - same as void
fn log(msg: &str) {
    println!("[LOG] {msg}");
}
// early return to guard against bad input
fn safe_div(a: f64, b: f64) → f64 {
    if b == 0.0 { return f64::NaN; }
    a / b
}
fn main() {
    println!("{}", square(4.0));           // 16
    log("started");
    println!("{}", safe_div(1.0, 0.0)); // NaN
}
```

Multiple values via tuple destructuring:

```
// return a tuple - idiomatic multi-value return
fn min_max(v: &[i32]) → (i32, i32) {
    let (mut lo, mut hi) = (v[0], v[0]);
    for &x in v {
        if x < lo { lo = x; }
        if x > hi { hi = x; }
    }
    (lo, hi) // return tuple
}
fn main() {
    let data = [3, 1, 4, 1, 5, 9, 2, 6];
    // destructure the tuple on the left
    let (lo, hi) = min_max(&data);
    println!("min={lo}, max={hi}"); // min=1, max=9
    // ignore one field with _
    let (_, hi) = min_max(&data);
    println!("max={hi}");           // max=9
}
```



Ownership rules

1. Each value in Rust has an **owner**.
2. There can only be **one owner** at a time.
3. When the owner goes out of scope, the value will be **dropped**.

Rules of References

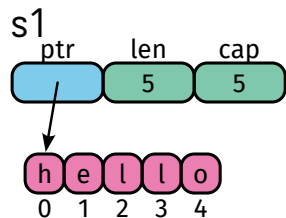
1. You can have **either** one mutable reference **or** any number of immutable references.
2. References must always be valid (no dangling references).

```
let s1 = String::from("hello");  
// Ownership moves to s2  
let s2 = s1;  
// println!("{s1}"); // (!) error: moved  
// Borrowing: many immutable refs OK  
let r1 = &s2;  
let r2 = &s2;  
println!("{r1} {r2}");  
// Or one mutable ref...  
let mut s3 = String::from("world");  
let m = &mut s3;  
m.push_str("!");  
// let r3 = &s3; // (!) error: already  
// borrowed as mutable  
println!("{m}");
```

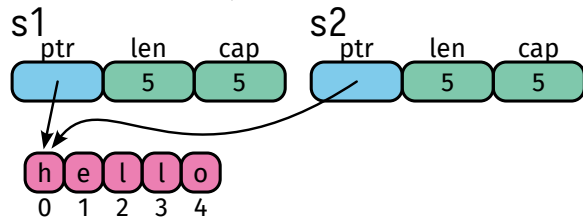


`&str` is a Copy type (a pointer to a string slice), while `String` is a Move type (a heap-allocated string):

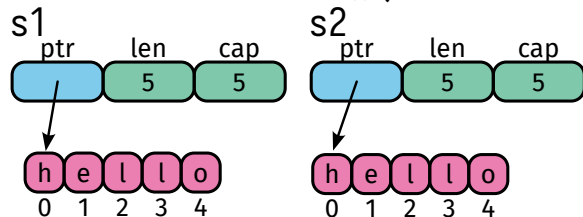
```
let s1 = "hello";
```



```
let s2 = s1;
```

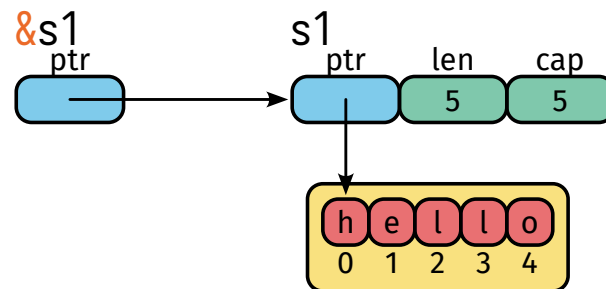


```
let s2 = s1.clone();
```



```
let s1 = String::from("hello");
```

```
&s1
```





Passing a value **transfers** or **copies** it depending on the type:

```
// i32 is Copy - the caller keeps its copy
fn double(n: i32) → i32 { n * 2 }

// String is NOT Copy - ownership moves into the fn
fn shout(s: String) → String {
    s.to_uppercase() // takes ownership, returns new value
}

fn main() {
    // Copy types: original still valid
    let x = 5;
    println!("{}", double(x)); // 10
    println!("{}", x);         // 5 ✓ still usable

    // Move types: original is gone after the call
    let name = String::from("alice");
    let loud = shout(name);    // name moved into shout()
    // println!("{}", name);    // ✕ compile error: moved!
    println!("{}", loud);     // ALICE
}
```

Copy types - value is duplicated	Move types - ownership transfers
bool, char	String
i8...i128, isize	Vec<T>
u8...u128, usize	HashMap<K, V>
f32, f64	Box<T>
(T, U) if all Copy	File, TcpStream
[T; N] if T: Copy	any struct by default
&T (shared ref)	&mut T (unique ref)



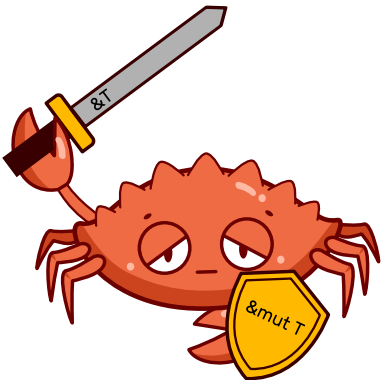
Borrow a value instead of taking ownership:

```
// &T - immutable borrow: read-only, never moves
fn length(s: &str) -> usize {
    s.len()           // caller keeps ownership
}

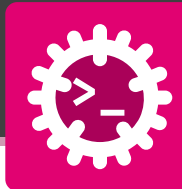
// &mut T - mutable borrow: read+write, still no move
fn double_all(v: &mut Vec<i32>) {
    for x in v.iter_mut() { *x *= 2; }
}

fn main() {
    let hello = String::from("hello");
    println!("{}", length(&hello)); // 5
    println!("{}", hello);           // ✓ still owned here

    let mut nums = vec![1, 2, 3];
    double_all(&mut nums);           // lend mutably
    println!("{}", nums);            // [2, 4, 6]
}
```



Syntax	Alias	Rule
T	move/copy	1 owner
&T	shared ref	many readers
&mut T	unique ref	1 writer



Anonymous functions

A closure is a function-like value written directly where it is needed. In **Python** and **C/C++**, they are called **lambda functions**.

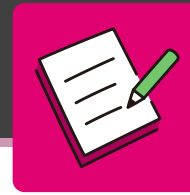
```
fn main() {  
    let square = |x: i32| x * x;  
    println!("{}", square(4)); // 16  
  
    let factor = 10;  
    let scale = |x: i32| x * factor;  
    println!("{}", scale(3)); // 30  
}
```

Closure syntax

```
|x| x + 1  
|x: i32| x * x  
|x: i32| → i32 {  
    x + 1  
}
```

Key idea

- Anonymous function
- Can be stored in a variable
- Can use values from the surrounding scope



Write **two small functions** for an exam grading system.

1. `swiss_grade(points: u32) → f64`

The exam has **5 exercises** of **0-10 points** each (max = 50). Convert the points to a **Swiss grade** (1-6):

$$\text{grade} = \frac{5}{\text{points}_{\text{max}}} * \text{points}_{\text{student}} + 1 \quad (1)$$

2. `bologna_grade(grade: f64) → char`

Map the Swiss grade to a **Bologna letter** (see table).

3. In main, print points, Swiss grade (2 decimals) and the letter for 50, 45, 40, 35, 30, 25 points.

Letter	Swiss grade
A	5.75 <= 6.00
B	5.25 <= 5.74
C	4.75 <= 5.24
D	4.25 <= 4.74
E	3.75 <= 4.24
F	1.00 <= 3.74



Use as `f64` to mix `u32` points with floating-point maths.



User-Defined Types

Let's make it objects!



One type, several variants

An enum defines a type that can be **one of several alternatives**.

```
enum TrafficLight {  
    Red,  
    Yellow,  
    Green,  
}  
  
fn main() {  
    let light = TrafficLight::Green;  
    match light {  
        TrafficLight::Red    => println!("stop"),  
        TrafficLight::Yellow => println!("prepare"),  
        TrafficLight::Green  => println!("go"),  
    }  
}
```

Key idea

- A value has exactly **one** variant
- Variants belong to the enum type
- **match** is used to handle all cases
- The compiler checks completeness



A variant can store values

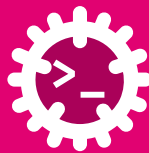
```
enum SensorValue {  
    Temperature(f64),  
    Pressure(f64),  
    Humidity(f64),  
}  
  
fn main() {  
    let value = SensorValue::Temperature(23.7);  
    match value {  
        SensorValue::Temperature(t) => println!("{t} °C"),  
        SensorValue::Pressure(p)    => println!("{p} Pa"),  
        SensorValue::Humidity(h)    => println!("{h} %"),  
    }  
}
```

Pattern matching extracts the value

```
SensorValue::Temperature(t)  
                        ^  
                        |  
                    extracted value
```



`Option<T>` and `Result<T,E>` are simply enums whose variants also contain data.



Value or no value

`Option<T>` is used when a value may be absent.

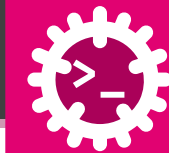
```
fn first(values: &[i32]) → Option<i32> {
    if values.len() > 0 {
        Some(values[0])
    } else {
        None
    }
}

fn main() {
    let data = [10, 20, 30];
    match first(&data) {
        Some(v) ⇒ println!("first = {v}"),
        None    ⇒ println!("empty list"),
    }
}
```

```
enum Option<T> {
    Some(T),
    None,
}
```

Meaning

- `Some(T)` contains a value
- `None` means no value
- Rust has no `null`

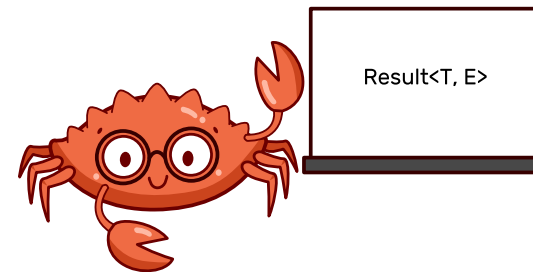


Success or error

`Result<T, E>` is used when an operation can fail.

```
fn parse_port(s: &str) → Result<u16, String> {
    match s.parse::<u16>() {
        Ok(port) ⇒ Ok(port),
        Err(_)   ⇒ Err(String::from("invalid port")),
    }
}

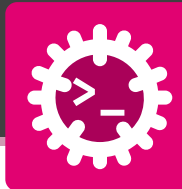
fn main() {
    let result = parse_port("8080");
    match result {
        Ok(port) ⇒ println!("port = {port}"),
        Err(msg) ⇒ println!("error: {msg}"),
    }
}
```



```
enum Result<T, E> {
    Ok(T),
    Err(E),
}
```

Meaning

- `Ok(T)` contains the successful value
- `Err(E)` contains the error value
- The caller must handle both cases



unwrap() was a mistake

Sometimes we are sure that an `Option` contains `Some(T)` or a `Result` contains `Ok(T)`. `unwrap()` extracts the value directly.



Option<T>

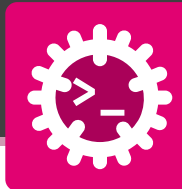
```
fn main() {  
    let port: Option<u16> = Some(8080);  
    let value = port.unwrap();  
    println!("port = {value}"); // 8080  
}
```

```
fn main() {  
    let port: Option<u16> = None;  
    let value = port.unwrap(); // panic  
    println!("port = {value}");  
}
```

Result<T, E>

```
fn main() {  
    let port: Result<u16, &str> = Ok(8080);  
    let value = port.unwrap();  
    println!("port = {value}"); // 8080  
}
```

```
fn main() {  
    let port: Result<u16, &str> = Err("invalid port");  
    let value = port.unwrap(); // panic  
    println!("port = {value}");  
}
```

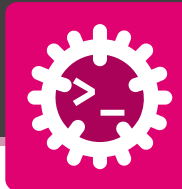


`unwrap()` was a mistake



i `expect("message")` works like `unwrap()`, but adds your own panic message.

```
fn main() {  
    let port: u16 = "8080".parse::<u16>().expect("port must be a number between 0 and 65535");  
}  
  
fn main() {  
    let port: u16 = "8080"  
        .parse::<u16>()  
        .expect("port must be a number between 0 and 65535");  
}
```



`unwrap()` was a mistake

Useful in examples, dangerous in real applications

`unwrap()` and `expect()` stop the program when the value is `None` or `Err`.



Better error message with `expect()`

Acceptable in small examples or tests

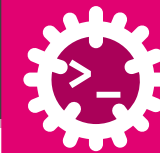
```
#[test]
fn parses_valid_port() {
    let port: u16 = "8080".parse::<u16>().unwrap();
    assert_eq!(port, 8080);
}
```

```
fn main() {
    let config = std::fs::read_to_string("config.toml")
        .expect("config.toml must exist next to the program");
    println!("{config}");
}
```



Use `expect()` only when crashing is acceptable and the message helps debugging.

Be Careful with `unwrap()` and `expect()`



`unwrap()` was a mistake

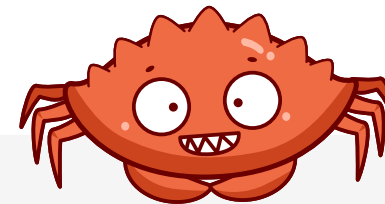
Better: handle the error

```
fn main() {
    let text: &str = "42";
    // unwrap: give me the value, or PANIC
    // parse() returns Result<u32, ParseIntError>
    let n: u32 = text.parse().unwrap();
    println!("{n}");           // 42

    // expect: same, but with a message
    let m: u32 = text.parse().expect("not a
number");
    println!("{m}");           // 42

    // if text were "abc", BOTH would crash:
    // thread 'main' panicked: ParseIntError ...
}
```

Better: provide a safe fallback



```
fn main() {
    let text = "abc";
    // match: handle BOTH outcomes explicitly
    match text.parse::<u32>() {
        Ok(n) => println!("got {n}"),
        Err(_) => println!("not a number"),
    }

    // if let: act ONLY on success, skip the rest
    if let Ok(n) = text.parse::<u32>() {
        println!("got {n}");
    }

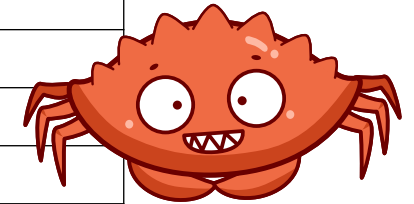
    // unwrap_or: supply a fallback, no crash
    let n = text.parse::<u32>().unwrap_or(0);
    println!("n = {n}");       // n = 0
}
```

Be Careful with `unwrap()` and `expect()`

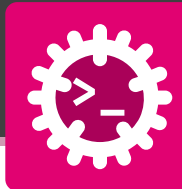


`unwrap()` was a mistake

Method	For	Behaviour
<code>unwrap()</code>	Option/Result	extract value or panic
<code>expect(msg)</code>	Option/Result	panic with custom message
<code>unwrap_or(default)</code>	Option/Result	extract value or use default
<code>unwrap_or_else(f)</code>	Option/Result	extract value or call closure
<code>unwrap_or_default()</code>	Option/Result	extract value or <code>Default::default()</code>
<code>ok_or(err)</code>	Option	convert to Result with error
<code>ok_or_else(f)</code>	Option	convert to Result with closure error
<code>unwrap_err()</code>	Result	extract error or panic
<code>expect_err(msg)</code>	Result	panic with msg if not Err
<code>match</code>	Option/Result	Handle success and failure explicitly
<code>?</code>	Option/Result	Return the error to the caller



`unwrap()` is convenient for examples and quick prototypes, but can crash the program if the value is `None`. Use it with caution.



Named fields

A struct groups related values into a single custom type.

```
#[derive(Debug)]
struct Sensor {
    id: u32,
    value: f64,
    unit: String,
}

fn main() {
    let s = Sensor {
        id: 1,
        value: 23.7,
        unit: String::from("°C"),
    };

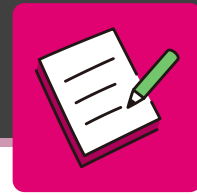
    println!("{:?}", s);
    println!("Sensor {}: {} {}", s.id, s.value, s.unit);
}
```

```
Sensor { id: 1, value: 23.7, unit: "°C" }
Sensor 1: 23.7 °C
```

Key idea

- Create your own type
- Group related data together
- Fields have names and types
- Access fields with .
- Improves readability over tuples

```
sensor.id
sensor.value
sensor.unit
```



Define a small **struct** and write one `Option` and one `Result` function:

1. `Struct Item { name: String, stock: u32 }`.
2. `take(item: &Item) → Option<u32>` - `Some(stock - 1)` if `stock > 0`, otherwise `None`.
3. `parse_qty(s: &str) → Result<u32, String>` - `Ok(n)` if `s` parses as a `u32`, otherwise `Err("not a number")`.
4. In `main`, create an `Item` and print the result of each function.



Reach for `Option` when a value **might be missing**, and `Result` when it **might fail**.



Ownership and Borrowing

And what the hell are Lifetimes 'a?



Every value has one owner

In Rust, every value has an **owner**. When the owner goes out of scope, the value is dropped.

```
fn main() {  
    {  
        let name = String::from("Ferris");  
        println!("{name}");  
    } // name goes out of scope here  
    println!("{name}"); // error  
}
```

Ownership rules

1. Each value in Rust has exactly one owner.
2. There can only be one owner of a value at any given time.
3. When the owner goes out of scope, the value is dropped



Ownership is Rust's way to manage memory safely at compile time.



Ownership can be transferred

Values like String are moved when assigned to another variable.

```
fn main() {  
    let a = String::from("Ferris");  
    let b = a; // ownership moves to b  
    println!("{b}");  
    println!("{a}"); // error  
}
```

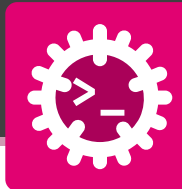
What happens?

```
let b = a;
```

- a no longer owns the string
- b is now the owner
- The data is not copied
- This prevents double-free errors



After a move, the old variable can no longer be used.



Some values are copied instead of moved

Simple values like integers, floats, booleans, and characters are copied.

```
fn main() {  
    let a = 42_u8;  
    let b = a; // copy  
    println!("a = {a}");  
    println!("b = {b}");  
}
```

```
fn main() {  
    let p = (10_u8, 20_u128);  
    let q = p;  
    println!("{:?}", p);  
    println!("{:?}", q);  
}
```

Copy types

- i32, u32, f64
- bool
- char
- Tuples containing only Copy types



Copy values are small and simple.
Rust duplicates them
automatically.



Function calls can move ownership

Passing a non-Copy value to a function transfers ownership.

```
fn print_name(name: String) {  
    println!("{name}");  
}  
fn main() {  
    let name = String::from("Ferris");  
    print_name(name);  
    println!("{name}"); // error  
}
```

Why?

- name is moved into print_name
- The function becomes the owner
- At the end of the function, the value is dropped



If a function does not need ownership, pass a reference instead.



Read without taking ownership

A reference lets a function use a value without owning it.

```
fn print_name(name: &String) {  
    println!("{name}");  
}  
fn main() {  
    let name = String::from("Ferris");  
    print_name(&name);  
    println!("{name}"); // still valid  
}
```

Borrowing

- `&String` is a reference
- The function can read the value
- Ownership stays in `main`
- The value is not dropped by the function

```
print_name(&name);  
//           ^  
//           borrow
```



Change without taking ownership

A mutable reference allows a function to modify a value.

```
fn add_unit(text: &mut String) {
    text.push_str(" °C");
}
fn main() {
    let mut value = String::from("23.7");
    add_unit(&mut value);
    println!("{value}");
}
```

Rules

- The variable must be declared `mut`
- The reference must be `&mut`
- Only one mutable reference is allowed at a time

```
let mut value = String::from("23.7");
add_unit(&mut value);
```



Mutable borrowing allows controlled modification without losing ownership.



Many readers or one writer

This prevents data races and invalid references at compile time.

Allowed: many immutable borrows

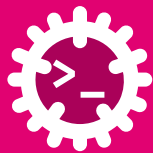
```
fn main() {  
    let value = String::from("23.7 °C");  
    let a = &value;  
    let b = &value;  
    println!("{a}");  
    println!("{b}");  
}
```

Not allowed: read and write at same time

```
fn main() {  
    let mut value = String::from("23.7");  
    let read = &value;  
    let write = &mut value; // error  
    println!("{read}");  
    println!("{write}");  
}
```



Rule of thumb: either many readers or exactly one writer.



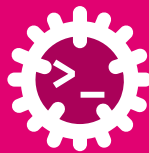
The compiler tracks how long references are valid

A **lifetime** is the span for which a reference stays valid. The **borrow checker** rejects any reference that could outlive the data it points to - so no dangling pointers and no use-after-free, all without a garbage collector.

× Dangling reference - rejected at compile time

```
fn main() {  
    let r;  
    {  
        let value = String::from("Ferris");  
        r = &value;           // r borrows value  
    }                         // value dropped here!  
    println!("{r}");          // r would read freed memory  
}
```

```
error[E0597]: `value` does not live long enough  
    r = &value;  
      ^^^^^ borrowed value does not live long  
enough  
    } - `value` dropped here while still borrowed  
    println!("{r}");  
      — borrow later used here
```



✓ Valid - the reference never outlives its owner

```
fn main() {  
    let value = String::from("Ferris"); //  
    let r = &value; // value  
    println!("{r}"); // r  
} // both end
```

Key idea

- A reference must **not outlive** the value it borrows
- Checked at **compile time only** - zero runtime cost
- Usually **inferred**; explicit 'a syntax shown next



Lifetimes don't change **when** a value is dropped - they let the compiler **prove** every reference is valid for as long as it is used.



Naming a lifetime so the compiler can relate references

Lifetimes are usually **inferred**. You only name one when a function **returns a reference** and the compiler cannot tell which input it borrows from.

```
// 'a links the inputs to the output:  
// "the result lives as long as both inputs do"  
fn longest<'a>(in1: &'a str, in2: &'a str) -> &'a str {  
    if in1.len() > in2.len() { in1 } else { in2 }  
}  
  
fn main() {  
    let s1 = String::from("Ferris");  
    let s2 = String::from("Rust");  
    println!("{}", longest(&s1, &s2)); // Ferris  
}
```

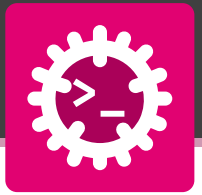
Reading the signature

- <'a> - declare a lifetime named a
- &'a str - “a reference valid for 'a”
- inputs and output **share** 'a, so the returned reference cannot outlive in1 or in2

Think of 'a as a label that links references together



Rule of thumb: if a function takes references and returns one, it usually needs a lifetime to say **which** input the result borrows from.



Each snippet below **does not compile**. Find the **ownership / borrowing problem** and say how you would fix it.

A

```
let s = String::from("hi");
let t = s;
println!("{s} and {t}");
```

B

```
let mut v = vec![1, 2, 3];
let first = &v[0];
v.push(4);
println!("first = {first}");
```

C

```
fn first_word() → &str {
    let s = String::from("hello world");
    s.split(' ').next().unwrap()
}
```

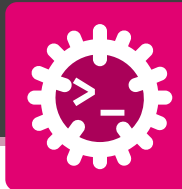


Think about **who owns** the value, **how long** a reference stays valid, and the rule **many readers XOR one writer**.



Composite Types

Let's mix and match!



Grouping several values - Composite types store several values together.

Type	Use case
[T; N]	Fixed-size array
Vec<T>	Growable list
HashMap<K, V>	Fast key-value lookup
BTreeMap<K, V>	Sorted key-value lookup
HashSet<T>	Unique values, fast lookup
BTreeSet<T>	Unique values, sorted
VecDeque<T>	First In First Out (FIFO) queue



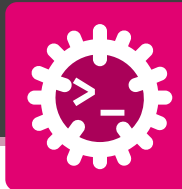
Most collection types are generic i.e. support any type.

```
use std::collections::{
    HashMap,
    HashSet,
    BTreeMap,
    BTreeSet,
    VecDeque,
};

fn main() {
    let numbers = [1, 2, 3];           // array
    let names = vec!["Ana", "Bob"];    // vector

    let mut ages = HashMap::new();     // HashMap
    ages.insert("Ana", 21);

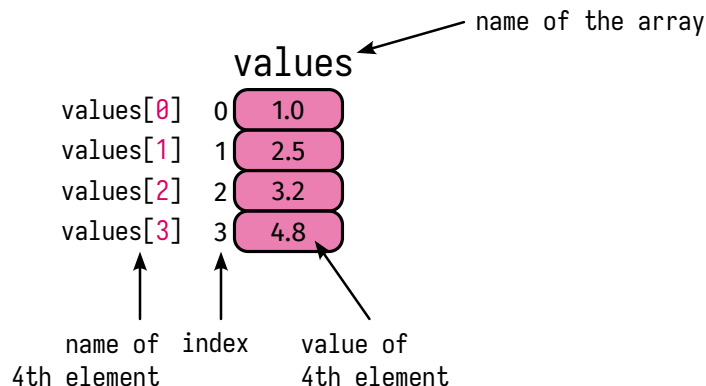
    let mut queue = VecDeque::new();   // VecDeque
    queue.push_back("task 1");
    queue.push_back("task 2");
}
```



[T; N] Fixed size, same type - An array stores a fixed number of values of the same type.

```
fn main() {  
    let values: [f32; 4] = [1.0, 2.5, 3.2, 4.8];  
    println!("length = {}", values.len()); // 4  
    println!("first = {}", values[0]);     // 1  
    println!("last  = {}", values[3]);     // 4.8  
    for value in values {  
        println!("{}", value); // 1  
    }                               // 2.5  
    }                               // 3.2  
    }                               // 4.8
```

```
[i32; 4]  
  |  |  
  |  +-- length  
  +----- element type
```



Array properties

- Fixed length
- Length is part of the type
- Stored contiguously
- Fast indexed access
- Useful for small fixed data

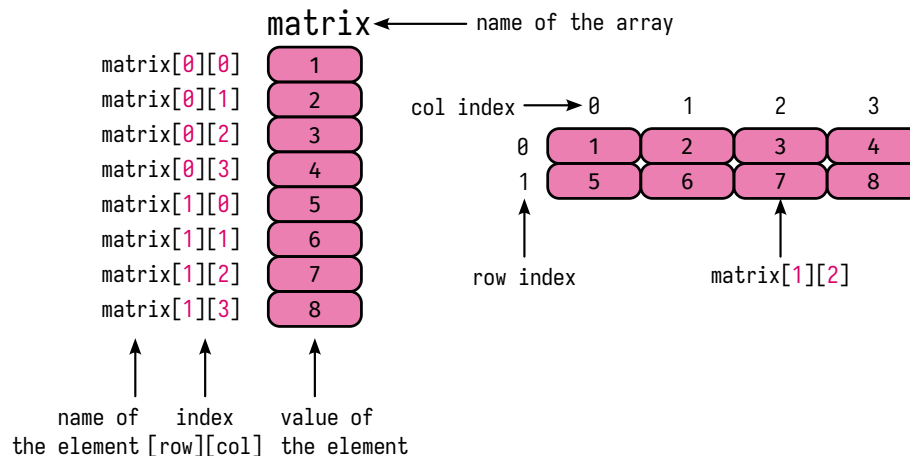


`[[T; N]; M]` Arrays can contain arrays - A multidimensional array is an array whose elements are also arrays.

```
fn main() {  
    let image: [[u8; 3]; 2] = [  
        [255, 128, 0],  
        [10, 20, 30],  
    ];  
    println!("{}", image[0][0]); // 255  
    println!("{}", image[1][2]); // 30  
  
    for row in image {  
        for pixel in row {  
            print!("{pixel:3} "); // 255 128 0  
        }                       // 10 20 30  
        println();  
    }  
}
```

`[[u8; 3]; 2]`
└── rows
└── columns
└── element type

```
let matrix: [[i32; 4]; 2] = [[1, 2, 3, 4], [5, 6, 7, 8]];
```



Typical use cases

- Small matrices
- Lookup tables
- Image kernels
- Fixed sensor grids

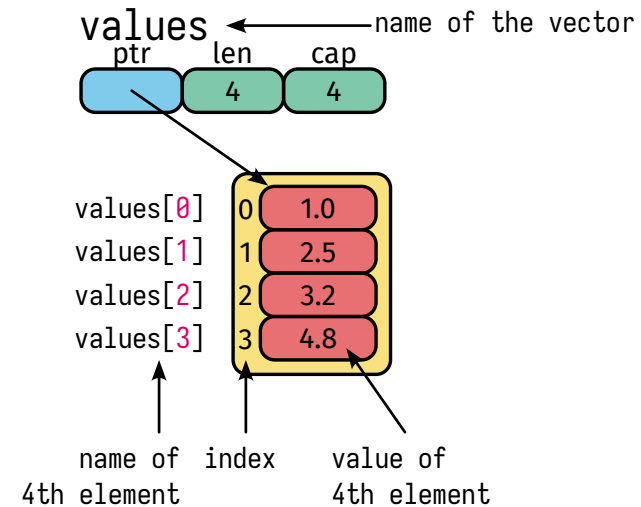


Vec<T> Growable list - A vector stores values of the same type and can grow at runtime.

```
fn main() {  
    let mut readings: Vec<f64> = Vec::new();  
  
    readings.push(23.7);  
    readings.push(24.1);  
    readings.push(23.9);  
  
    println!("len = {}", readings.len());  
    println!("first = {}", readings[0]);  
  
    for value in &readings {  
        println!("{value} °C");  
    }  
}
```

```
let mut values = Vec::new();  
values.push(10);  
values.push(20);
```

```
let values: Vec<f32> = vec![1.0, 2.5, 3.2, 4.8]
```



Vector properties

- Dynamic length
- Fast access by index
- Same element type
- Can add values with push



Safe Access with `get()`, avoiding index panics.

Indexing with `[i]` panics if the index is invalid. `get(i)` returns an `Option`.

```
fn main() {
    let readings = vec![23.7, 24.1, 23.9];
    // Direct indexing
    println!("{}", readings[1]); // 24.1
    // Safe access
    match readings.get(10) {
        Some(value) => println!("{value}"),
        None => println!("no such reading"),
    }
}
```

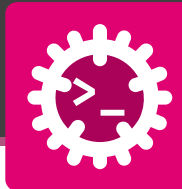
Why Option?

```
let value: Option<f64> = readings.get(10)
```

- `Some(value)` if the index exists
- `None` if the index is invalid



Prefer `get()` when the index may come from user input, files, sensors, or network data.



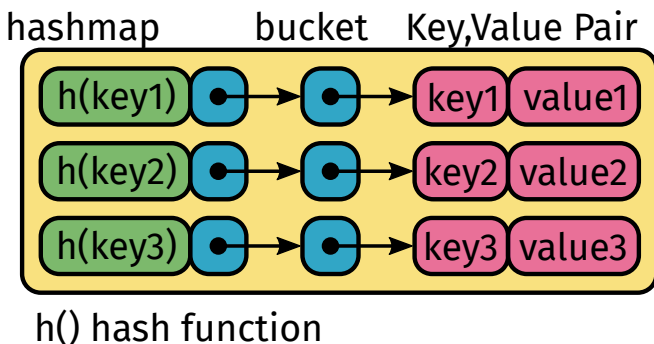
HashMap<K, V> Fast key-value lookup - Stores values under keys.

```
use std::collections::HashMap;

fn main() {
    let mut sensors = HashMap::new();
    sensors.insert("T1", 23.7);
    sensors.insert("T2", 24.1);
    sensors.insert("P1", 1013.2);
    match sensors.get("T1") {
        Some(value) => println!("T1 = {value}"),
        None => println!("sensor not found"),
    }
    for (id, value) in &sensors {
        println!("{id}: {value}");
    }
}
```

```
HashMap<String, f64>
      |           |
      |           | value type
      |           | key type
```

HashMap<K,V>

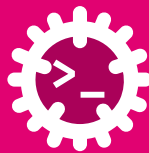


Typical use cases

- Sensor id to value
- Lookup tables
- User id to user data
- Configuration values
- Counting occurrences



- A HashMap is fast, but iteration order is not sorted.
- A bucket is a storage slot in a hash map.



A common real-world pattern - `entry()` is useful when a key may or may not already exist.

```
use std::collections::HashMap;

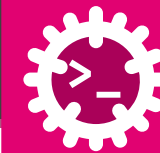
fn main() {
    let events = [
        "open", "close", "open",
        "error", "open", "error",
    ];
    let mut counts = HashMap::new(); // new empty HashMap
    for event in events {
        // if the key doesn't exist, insert it with value 0,
        // then get back a mutable reference to the value
        let counter = counts.entry(event).or_insert(0);
        // this modifies the value inside the HashMap
        *counter += 1;
    }
    println!("{counts:?}");
}
```

```
{
  "open": 3,
  "close": 1,
  "error": 2
}
```

Meaning

```
counts.entry(event).or_insert(0)
```

- Get existing counter
- Or insert 0
- Then increment it via `&mut`



BTreeMap<K, V> Sorted key-value lookup - stores key-value pairs sorted by key.

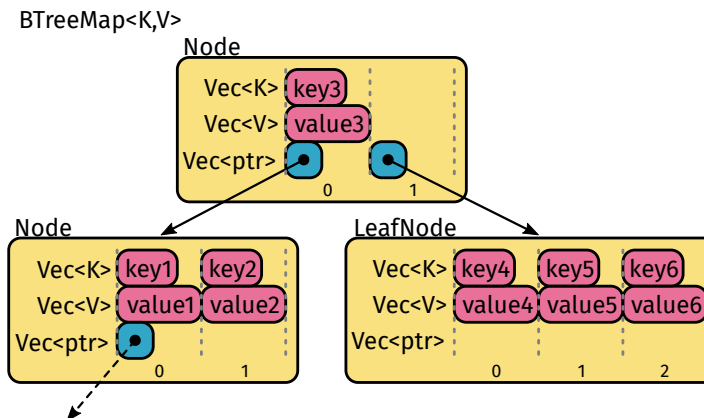
```
use std::collections::BTreeMap;

fn main() {
    let mut measurements = BTreeMap::new();

    measurements.insert("10:00", 23.7);
    measurements.insert("09:00", 22.9);
    measurements.insert("11:00", 24.1);

    for (time, value) in &measurements {
        println!("{time}: {value}");
    }
}
```

```
09:00: 22.9
10:00: 23.7
11:00: 24.1
```

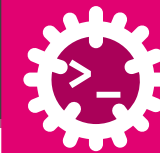


Use BTreeMap when you need

- Sorted keys
- Ordered iteration
- Range queries
- Predictable output order



Use HashMap for fast lookup. Use BTreeMap for ordered data.



Unique values - A HashSet<T> stores each value only once.

```
use std::collections::HashSet;

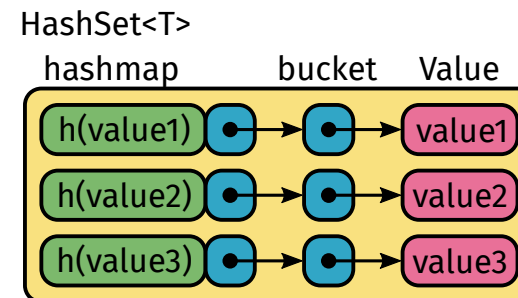
fn main() {
    let mut ids = HashSet::new();

    ids.insert("T1");
    ids.insert("T2");
    ids.insert("T1"); // duplicate

    println!("count = {}", ids.len());

    if ids.contains("T2") {
        println!("T2 exists");
    }
}
```

```
count = 2
T2 exists
```

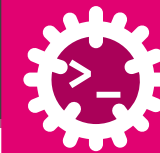


Typical use cases

- Remove duplicates
- Track seen elements
- Membership tests
- Compare groups



A HashSet<T> is basically a HashMap<T, ()> without values.



Unique and sorted values - A BTreeSet<T> stores unique values in sorted order.

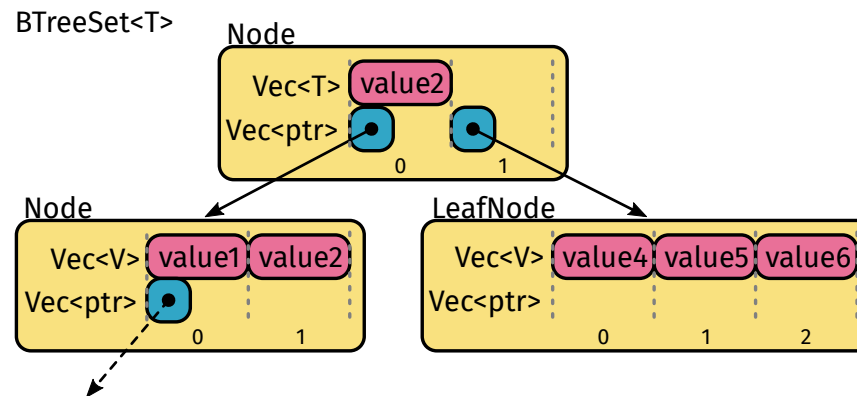
```
use std::collections::BTreeSet;

fn main() {
    let mut ids = BTreeSet::new();

    ids.insert("T3");
    ids.insert("T1");
    ids.insert("T2");
    ids.insert("T1");

    for id in &ids {
        println!("{}", id);
    }
}
```

```
T1
T2
T3
```



Use BTreeSet when you need

- Unique values
- Sorted output
- Deterministic iteration
- Range-like operations



Use HashSet for speed. Use BTreeSet for order.



First in, first out - A queue processes values in the same order as they arrive.

```
use std::collections::VecDeque;

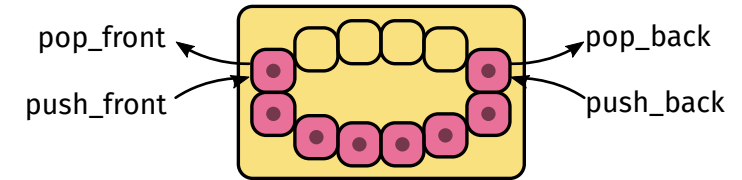
fn main() {
    let mut queue = VecDeque::new();

    queue.push_back("read sensor");
    queue.push_back("filter value");
    queue.push_back("send packet");

    while let Some(task) = queue.pop_front() {
        println!("processing: {task}");
    }
}
```

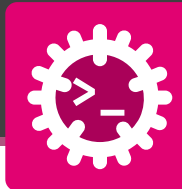
```
processing: read sensor
processing: filter value
processing: send packet
```

Double Ended Queue VecDeque<T>



Queue operations

push_front(x)	add at the front
push_back(x)	add at the end
pop_front()	remove from the front
pop_back()	remove from the back
front()	read first item
back()	read last item

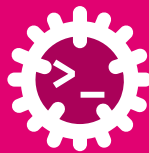


Practical rule of thumb

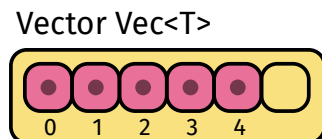
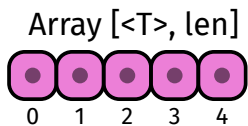
Need	Use
Fixed number of values	Array [T; N]
Growable list	Vector Vec<T>
Fast lookup by key	HashMap<K, V>
Sorted lookup by key	BTreeMap<K, V>
Unique values	HashSet<T>
Unique sorted values	BTreeSet<T>
FIFO queue	VecDeque<T>



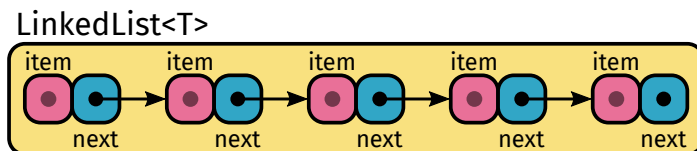
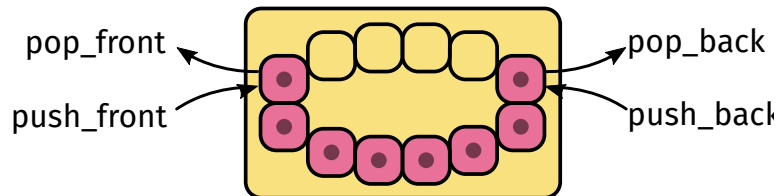
Start with Vec<T>. Use maps, sets, and queues when the access pattern requires them.



Sequences

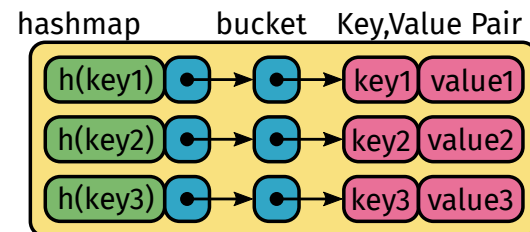


Double Ended Queue $\text{VecDeque}<T>$



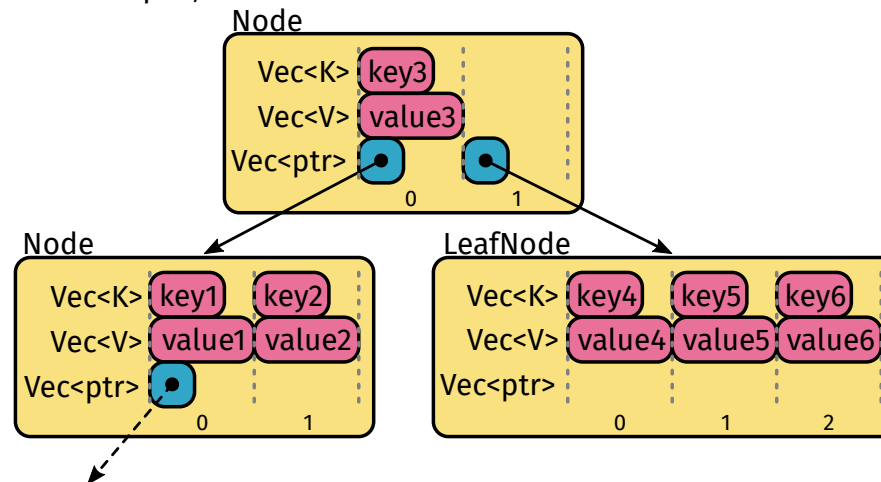
Maps

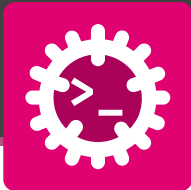
HashMap $<K,V>$



$h()$ hash function

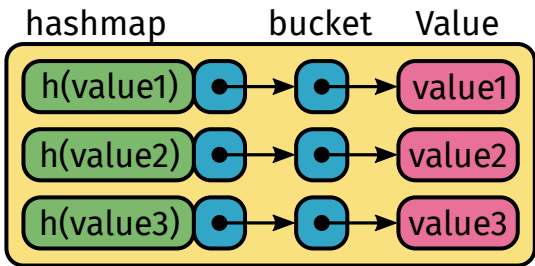
BTreeMap $<K,V>$



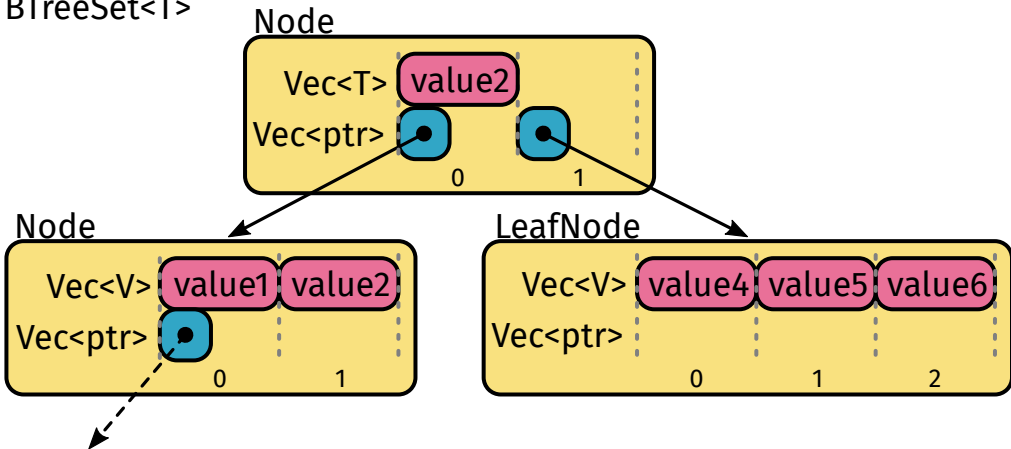


Sets

HashSet<T>

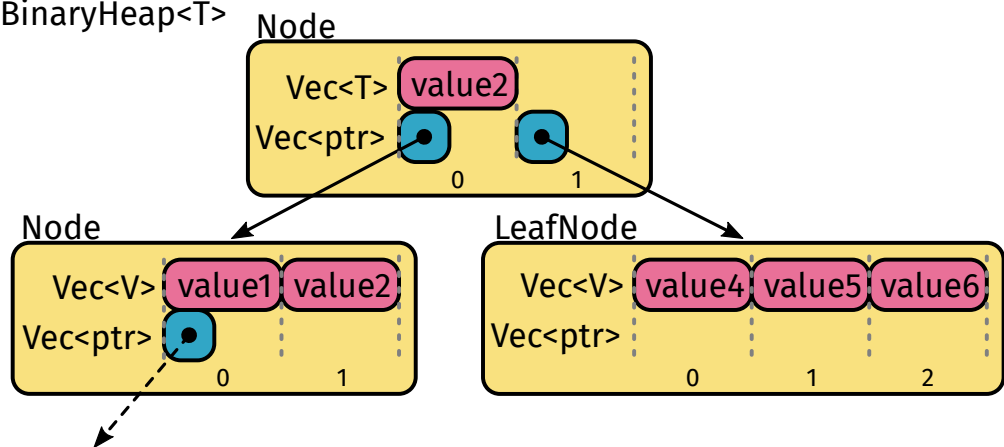


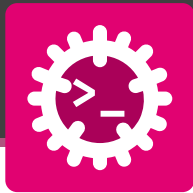
BTreeSet<T>



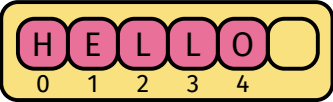
🔗 Misc

BinaryHeap<T>

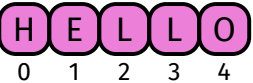




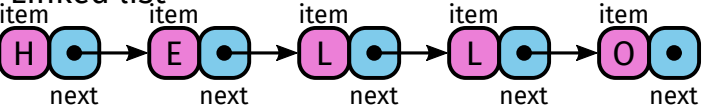
Vector



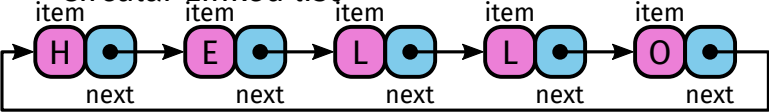
Array



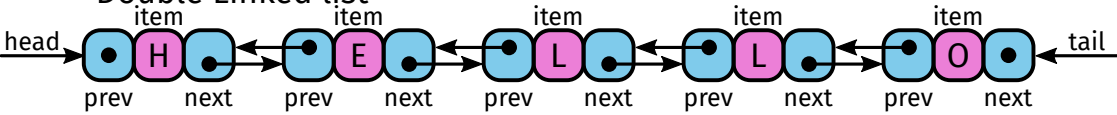
Linked list



Circular Linked list

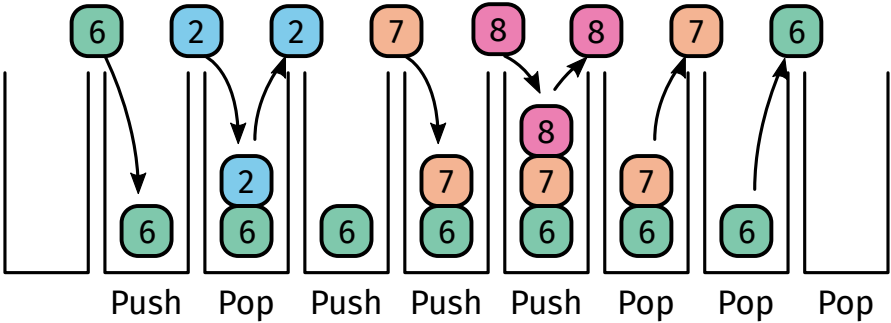


Double Linked list



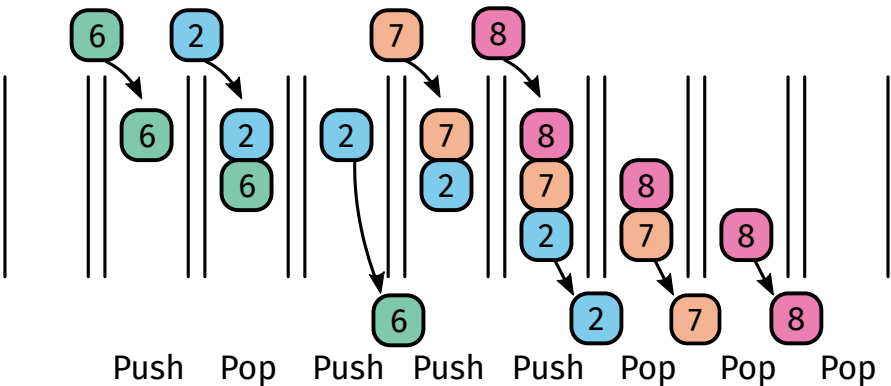
Stack

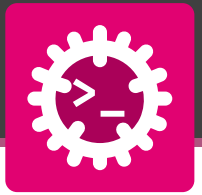
LIFO - Last In First Out



Queue

FIFO - First In First Out





Read the program and write the value printed for a, b, c and d.

```
use std::collections::HashMap;

fn main() {
    let votes = vec!["red", "blue", "red",
                     "green", "red", "blue"];
    let total = votes.len();

    let mut tally: HashMap<&str, u32> = HashMap::new();
    for v in votes {
        *tally.entry(v).or_insert(0) += 1;
    }

    println!("a {}", total);           // a
    println!("b {}", tally.len());     // b
    println!("c {}", tally["red"]);    // c
    println!("d {:?}", tally.get("yellow")); // d
}
```



`entry(k).or_insert(0)` returns a `&mut` to the value, inserting 0 first if the key is new.

`tally.get(k)` returns an `Option`, while `tally[k]` panics on a missing key.

WHY DO WHALES JUMP
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY WUBA LUBBA DUB DUB MEANING
WHY IS THERE A WHALE AND A POT FALLING
WHY ARE THERE SO MANY BIRDS IN SWISS
WHY IS THERE SO LITTLE RAIN IN WALLIS
WHY IS WALLIS WEATHER FORECAST ALWAYS WRONG

WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
HOW FAST IS LIGHTSPEED
WHY ARE OLD KLINGONS DIFFERENT

WHY ARE THERE
SQUIRRELS



WHY AREN'T THERE DINOSAUR GHOSTS
WHY DO IGUANAS DIE
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY HAVE DINOSAURS NO FUR
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY IS 42 THE ANSWER TO EVERYTHING
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER
WHY IS MARVIN ALWAYS SO SAD

WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY IS 42 THE ANSWER TO EVERYTHING
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER
WHY IS MARVIN ALWAYS SO SAD

WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN

WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPYDER BITES ITCH

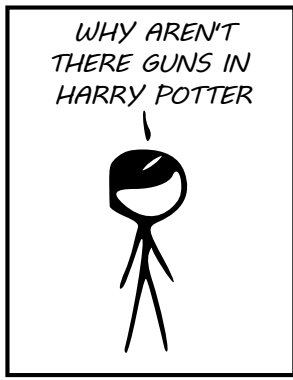
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY HAVE DINOSAURS NO FUR
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY IS 42 THE ANSWER TO EVERYTHING
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER
WHY IS MARVIN ALWAYS SO SAD

QUESTIONS

CAN BE ASKED BY ANYONE ANYTIME

WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY IS 42 THE ANSWER TO EVERYTHING
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER
WHY IS MARVIN ALWAYS SO SAD

WHY ARE THERE FEMALE



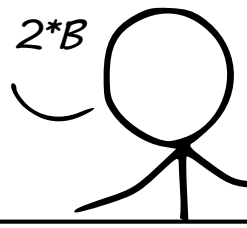
WHY IS THERE A SWARM OF ANTS
WHY IS THERE PILGRIM
WHY ARE THERE SO MANY CROWS IN ROCHESTER
WHY IS TO BE OR NOT TO BE FL
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE

WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE FPGA's EVERYWHERE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOCKS

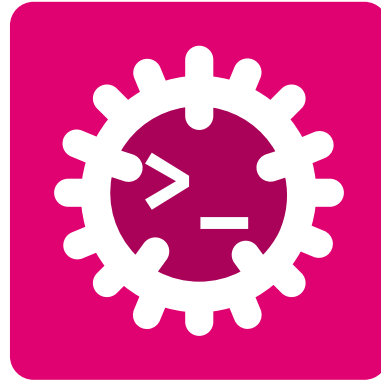
WHAT IS <https://xkcd.com/1256/>
WHY DO THEY SAY T-MINUS
WHY ARE OCEANS BECOMING MORE ACIDIC

WHY IS HTTPS CROSSED OUT
WHY IS THERE A LINE THROUGH
WHY IS THERE A RED LINE THROUGH HTTPS
WHY IS HTTPS IMPOR
WHY IS YKK ON ALL ZIPPERS

WHY 2*B / 2*B



WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO PEOPLE DIE



Glossary & Bibliography



Computer Science

ASCII – American Standard Code for Information Interchange: Character encoding standard for representing text. [29](#)

RGB – Red Green Blue: Color model representing colors as combinations of red, green, and blue. [29](#)

Data Structure

FIFO – First In First Out: Queue discipline where the oldest element is processed first. [105](#), [116](#)

Programming

Heap – Heap: Memory area used for dynamic memory allocation. [48](#)

Stack – Stack: Memory area used for static memory allocation and function call management. [13](#)

USV – Unicode Scalar Value: A Unicode code point that is not a surrogate and represents a single character. [46](#)

UTF – Unicode Transformation Format: Encoding standard for representing Unicode characters. [48](#)



Bibliography

- [1] S. Klabnik, C. Nichols, and C. Kryocho, “The Rust Programming Language.”
- [2] J. Blandy, J. Orendorff, and L. F. s Tindall, *Programming Rust: Fast, Safe Systems Development*. O'Reilly Media, Incorporated, 2021.
- [3] H. Collard, *Black Hat Rust*, vol. 1. Amazon, 2025.
- [4] A. Shvets, “Refactoring Guru.” 2026.
- [5] R. C. Martin, *Clean Code: A Handbook of Agile Software Craftsmanship*. Pearson Education, 2008.
- [6] “SVG Repo - Free SVG Vectors and Icons.”